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Second-year engineering internship

Predicting Which Antenna a Mobile Phone Will Use Next

A language model retrained on one day of network records

Olivier Recher

<i>Host institution</i>	Czech Technical University in Prague (ČVUT)
<i>Academic supervisor</i>	Robert Besták
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Abstract

Mobile terminals continuously attach to cellular antennas, with operators logging transitions as discrete event logs. This report evaluates whether parameter-efficient fine-tuned language models trained on anonymized call detail records can forecast a subscriber’s next cellular antenna attachment superior to a deterministic benchmark. Predictions are evaluated against the *persistence baseline* (“*the next cell matches the current cell*”), which achieves an accuracy of 69.2% on unfiltered daily logs due to subscriber stationary intervals and serves as the primary benchmark across all experimental evaluations.

The empirical dataset comprises fifteen daily logs from the Karlovy Vary study area, encompassing 369 cells mounted across 113 base stations. On a held-out test cohort of 1,000 subscribers, the fine-tuned model achieves a **+4.3 % accuracy gain** over the persistence baseline (54.5% vs. 50.2% top-1 accuracy). Surpassing the persistence baseline required conditioning parameter updates on *subscriber temporal mobility profiles*, stratifying users into behavioral cohorts and routing trajectories to specialized LoRA parameter adapters. Experimental results demonstrate that cohort scaling saturates at approximately 2,000 subscribers (+4.2 % over baseline), while individual subscriber mobility typology accounts for a 22.4 % accuracy spread across behavioral groups (compared to 0.4 % across dataset scale). Trajectory predictability is fundamentally governed by subscriber mobility patterns rather than training dataset volume.

Keywords: human mobility prediction, call detail records,¹ next-cell prediction, large language models, LoRA, QLoRA, behavioral clustering, trajectory scaling.

¹**CDR**, *call detail record*: an anonymized network event log containing subscriber identifier, event timestamp, and attached cell identifier, excluding communication content.

Executive Summary

Research Objective. Predict the next cellular antenna attachment for a mobile phone trajectory based exclusively on the historical sequence of observed cell events. Accurate next-cell prediction enables mobile network operators to optimize network capacity allocation and reduce handover connection drops.

Dataset Characterization. Evaluation uses fifteen daily logs from a Czech mobile operator covering the Karlovy Vary study area (12–26 March 2014), comprising 369 cells mounted on 113 base stations across 80,217 to 99,504 daily subscriber trajectories. Records contain anonymized subscriber identifiers, event timestamps, and cell IDs, excluding personal data or message content.

Benchmark Reference Baseline. Predictions are evaluated against the **persistence baseline** (*“the next cell is identical to the current cell”*). On raw daily logs, this baseline achieves 69.2% accuracy due to high subscriber stationary time. Model performance is reported exclusively as accuracy deltas relative to this baseline benchmark.

Model Architecture. A 500M-parameter language model (Qwen2.5-0.5B-Instruct) is fine-tuned using Quantized Low-Rank Adaptation (QLoRA) on single GPU instances, adapting discrete language representations to spatial transition modeling.

Experimental Chronology. **Table 1** summarizes the experimental progression across eleven iterations, detailing cohort characteristics, architectural modifications, and baseline accuracy deltas (**Gap** column). The **Rule** column, the baseline’s own score, moves from row to row, which is why the absolute scores may not be compared across rows: test subscriber filtering and evaluation protocols changed four times across project stages (for instance, shifting from grading the “last fifth” of a day to grading the harder “whole day” sequence lowers baseline accuracy from 45.5% to 43.1% on identical subscriber data), so cross-iteration comparisons are valid only on relative baseline deltas. Early iterations (1–3, 5–7) operated below baseline, Iteration 4 produced a transient gain (+1.1 %) resolved by subsequent filtering, and sustained baseline outperformance (+3.7 % to +4.3 %) was established from Iteration 8 onward with per-group LoRA adapter routing. Iteration 9 occupies two rows because two independent changes were tested that week; Appendix A details what each iteration did.

Table 1: The eleven iterations, on one axis

It.	People	What I changed that week	Graded on	Rule	Model	Gap
1	RAW DAY	built the pipeline end to end	last fifth	69.2 %	16.9 %	−52.3
2	RAW DAY	read the best saved copy	last fifth	69.2 %	64.4 %	−4.8
3	RAW DAY, 5,000 users	ten times more people	last fifth	81.6 %	76.0 %	−5.6
4	FILTERED DAY	dropped near-silent people	last fifth	79.0 %	80.1 %	+1.1
5	MOBILE-USER DAY	kept only people who move	last fifth	45.5 %	44.6 %	−0.9
6	MOBILE-USER DAY	gave the model the hour of day	last fifth	45.5 %	44.2 %	−1.3
7	MOBILE-USER DAY	named the kind of user in the text	whole day	43.1 %	41.3 %	−1.8
8	MOBILE-USER DAY	one set of numbers per kind of user	whole day	43.1 %	46.8 %	+3.7
9	2,000-USER DAY	1,900 training people	last fifth	48.2 %	52.4 %	+4.2
9	LEVELLED CORPUS	stretched every day to 200 records	last fifth	76.5 %	74.9 %	−1.6
10	ELEVEN-DAY SET	21,000 training people	last fifth	50.2 %	54.5 %	+4.3
11	ELEVEN-DAY SET	<i>no new training: why the curve climbs, then stops</i>				

Key Research Findings.

1. **Baseline Outperformance via User Stratification:** The fine-tuned language model achieves a **+4.3 % top-1 accuracy gain** over the persistence baseline on held-out evaluation cohorts (54.5% vs. 50.2% top-1 accuracy). Surpassing the persistence baseline required stratifying subscribers into temporal mobility profiles (*G0–G3*) and routing trajectories to per-group LoRA adapters (Chapter 7).
2. **Data Cohort Scaling Saturation at 2,000 Subscribers:** Accuracy gains saturate at approximately 2,000 subscribers (+4.2 % over baseline). Expanding training cohorts to 21,000 subscribers yields no statistically significant improvement (+0.04 %, $p > 0.05$) (Chapter 8).
3. **Dominance of Subscriber Typology over Data Scale:** Individual subscriber mobility typology accounts for a **22.4 % accuracy spread** across behavioral groups, exerting a 50× stronger influence on predictability than scaling dataset size beyond 2,000 subscribers (Chapters 5 and 8).

Methodological Insights. Experimental evaluations established key methodological principles: applying early stopping at peak validation accuracy prevents overfitting (+15.5 % accuracy recovery over final epoch checkpoints); standardizing context fraction split protocols prevents evaluation split artifacts; and benchmarking on active mobility cohorts ensures meaningful performance measurement (Chapter 4).

Acknowledgements

My first thanks go to Robert Besták, who supervised this internship and took the time, week after week, to explain the parts of the mobile-network side of the problem that no amount of reading would have given me: what a signalling record actually is, how cells and base stations relate to one another, and why a question that looks like plain sequence prediction is constrained by the way the network itself is built. That patience shaped the direction of the work, and most of the corrections that mattered came out of those conversations.

I am equally grateful to the small group of interns working on the same corpus, Nino, Tiphain, Benjamin and Reda, for their help throughout these two months. They were already several months into their own analyses when I arrived, and they explained the corpus, the conventions and the pitfalls to me rather than leaving me to rediscover them. Nino in particular went out of his way, repeatedly, to produce the extracts I needed with the tooling he had built for his own analysis, which spared me a great deal of work and made most of the experiments in this report possible at all. Tiphain and Benjamin discussed their move-or-stay framing and their prompting results openly, and two of the ideas I reused here are theirs. Working next to them made those two months both easier and considerably more interesting than they would otherwise have been.

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1 Introduction

1.1 Host Laboratory and Dataset Description

This report documents a second-year engineering research internship conducted from 29 June to 28 August 2026 at the Czech Technical University in Prague (ČVUT), under the supervision of Robert Besták. The laboratory holds a large dataset of anonymized mobile network records provided by a Czech mobile telecommunications operator, which serves as the primary dataset for several research projects conducted by the laboratory’s interns.

These records contain no personally identifiable information, such as subscriber names or phone numbers, nor message content. Each record line specifies an anonymized subscriber identifier, the temporal timestamp, and the sequence of cellular antennas to which the mobile terminal was attached over a 24-hour period. This anonymized corpus constitutes the empirical foundation of the work described in this report.

1.2 Research Objectives and Next-Cell Prediction Task

Mobile terminals continuously attach to the cellular antenna offering the highest signal quality. As users move, the terminal handover connection from one antenna to the next. The mobile operator logs an event line upon each cell transition, as well as periodically during stationary periods.

The central research objective of this internship is formulated as follows: *given the initial segment of a subscriber’s daily mobility trajectory, can the model accurately predict the next cellular antenna to which the device will connect?* Predicting cellular transitions enables mobile network operators to optimize resource allocation, anticipate local capacity demands, and reduce connection drop rates through proactive handover initialization.

A fundamental distinction must be established between physical infrastructure and logical coverage areas.¹ A **base station** corresponds to a physical mast housing multiple directional antennas. Each individual antenna defines a distinct **cell**. The geographical study area evaluated in this work comprises **369 cells** distributed across **113 base stations**, representing an average of three to four cells per physical site. The target of the predictive task is the specific cell rather than the base station, which presents a higher predictive complexity than coarse-grained spatial localization.

¹**Base station** (or *site/mast*): a physical infrastructure site. **Cell**: the spatial coverage area served by an individual directional antenna mounted on a base station.

1.3 Next-Cell Prediction via Fine-Tuned Language Models

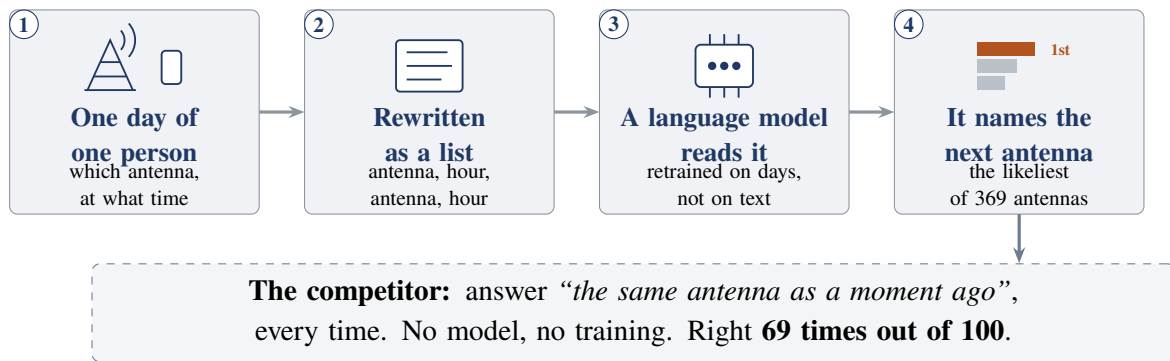


Figure 1.1: The four steps this internship built

As illustrated in **Figure 1.1**, the overall experimental architecture transforms continuous 24-hour attachment records into discrete token sequences, fine-tunes a language model on thousands of these daily subscriber traces, and lets the model output a ranked list of candidate cellular antennas out of 369 possibilities. The dashed benchmark box at the bottom of the figure is the persistence baseline, a deterministic competitor requiring no training that already predicts correctly roughly seven times out of ten; every performance metric in this report evaluates progress relative to it.

At the core of this architecture is a **language model**.² While language models are conventionally employed for natural language processing, their mathematical framework operates on generic ordered sequences of discrete tokens. In this context, daily mobility traces are framed as ordered sequences of cell identifiers selected from a fixed vocabulary of 369 discrete cells.

Rather than using off-the-shelf general language models directly, parameter-efficient **fine-tuning** (PEFT) is applied. The internal parameters of the neural network are adapted using local mobility traces, specializing the model from a general natural language generator into a dedicated next-cell predictor for the target geographical region, as detailed in Chapter 4. Alternative approaches explored in parallel research rely on prompting off-the-shelf models without parameter updates, addressing a distinct research scope.

1.4 Baseline Evaluation: The Persistence Baseline

To rigorously assess model performance, predictions must be evaluated against a strong deterministic benchmark: the **persistence baseline**.

The persistence baseline evaluates the heuristic rule: “the next cell is identical to the current cell”. Despite its simplicity, this single-line baseline achieves an accuracy of **69.2%** on raw daily network records, primarily because individuals remain stationary for large portions of the day. Beating it constitutes the key criterion for demonstrating true predictive capability, and across early experimental iterations it proved difficult to surpass.

²**Language model:** a probabilistic model trained to estimate the probability distribution over sequences of discrete tokens. The underlying mechanics operate on arbitrary discrete sequences, allowing its application to spatial mobility traces.

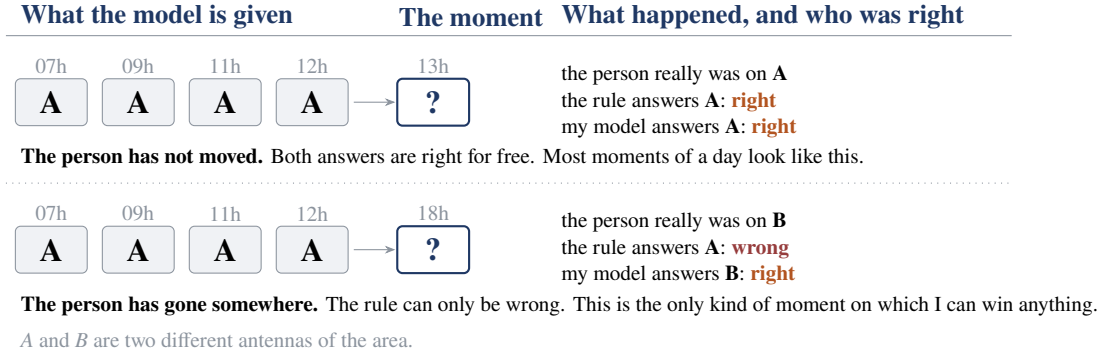


Figure 1.2: One prediction, taken apart

Figure 1.2 illustrates the mechanics of the persistence baseline compared to model predictions on two moments of the same subscriber’s day, one per row, where the letters A and B stand for two antennas. The model receives a historical trajectory prefix in which each moment is an hour and an antenna (left); at the evaluated moment, the hour is provided while the true target antenna is hidden (middle), and the predictions of the model and of the persistence rule are graded against ground truth (right). In the first row, the subscriber remains stationary, so repeating the previous antenna yields a correct prediction for both the baseline and the model without requiring predictive intelligence. In the second row, the subscriber moves to a new antenna and the persistence baseline necessarily fails. Transition events are therefore the only evaluation moments where a model can demonstrate genuine predictive gains over the baseline.

Throughout this report, performance gains are reported as the *difference in accuracy* relative to the persistence baseline. A reported gain of +4.3% indicates that the fine-tuned model achieves an accuracy of 54.5% on a benchmark where the persistence baseline yields 50.2% across identical evaluation instances. Evaluating relative performance delta ensures consistent comparisons across dataset revisions and experimental cohorts.

1.5 Related Work and Collaborative Context

This project was conducted alongside research by peer group members investigating the same mobile network dataset. **Figure 1.3** illustrates the placement of this study within the broader team research framework: all five researchers analyzed the shared 15-day mobile network corpus, shown at the top, across distinct research questions and methodological frameworks, and the four boxes across the middle are the other team members.

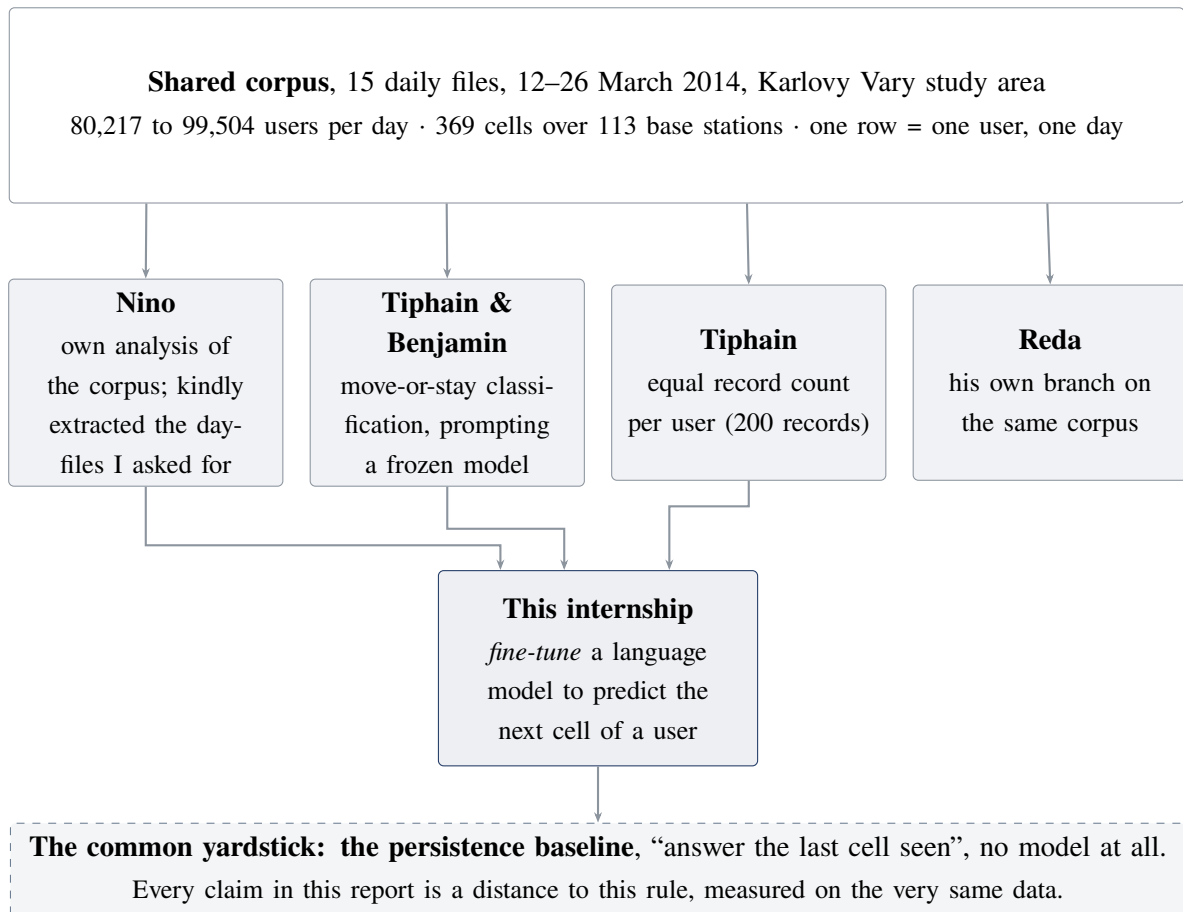


Figure 1.3: This internship inside the team’s work

Nino established automated corpus extraction scripts for identifying user home/activity cells and extracting daily mobility trajectories utilized in this study. Tiphain and Benjamin investigated zero-shot and few-shot prompting techniques on frozen LLMs alongside move-versus-stay classification. Tiphain also evaluated trajectory length equalization techniques (Chapter 8), while Reda conducted parallel mobility pattern analyses. The fine-tuned language model forecasting architecture established in this report sits at the bottom of **Figure 1.3**, and the dashed box of the same figure defines the common persistence baseline benchmark (“answer the last cell seen”) that ensures direct comparability across team findings.

1.6 Report Structure

This report is organized by thematic research questions rather than chronological execution; the complete timeline of experimental iterations is provided in Appendix A. Chapters 3 through 8 conclude with a synthesis section summarizing empirical conclusions. Appendix B details experimental hyperparameters and tabular data provenance, while Appendix C provides a reference glossary of technical domain terminology.

2 State of the Art

Eight published studies form the theoretical and empirical baseline for this investigation, all of which are cited in the bibliography. Two foundational papers established the theoretical bounds of trajectory predictability and guided the methodology of this study. Among the remaining six works, three establish standard benchmarking protocols and evaluation metrics for mobility predictors, two provide the theoretical basis for user stratification, and one defines the conceptual boundary between spatial location prediction and cellular antenna handovers.

2.1 Predictability Ceilings and Baseline Boundaries

The mobility literature provides critical bounds on theoretical predictability limits prior to model deployment. **Figure 2.1** places these bounds and the scores measured in this work on a single scale, the percentage of moments predicted correctly. The two upper grey bars are theoretical predictability ceilings published in the literature, the best score any predictor could reach; the two lower bars are measured on the evaluation datasets of this report, in blue for the persistence baseline and in green for the gain of the fine-tuned language model.

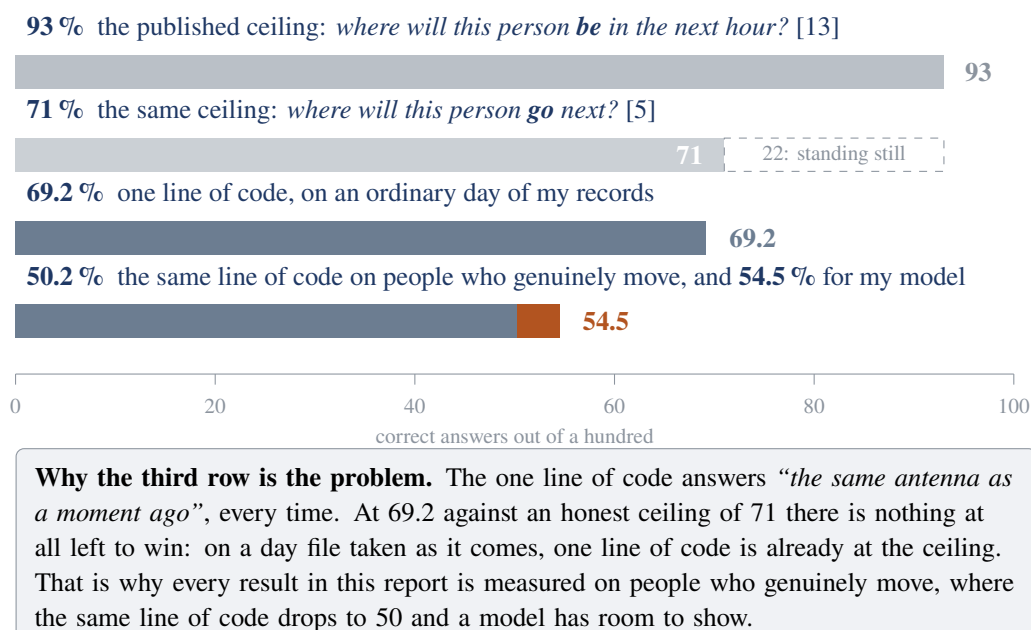


Figure 2.1: The room a predictor has, once standing still is taken out of it

Song et al. analyzed three months of mobile phone location records for 45,000 subscribers, modeling visited antenna sequences as discrete stochastic processes to quantify theoretical entropy bounds [13]. From information-theoretic entropy measurements, they derived an upper bound of 93% for overall location predictability, the first bar of **Figure 2.1**.

Subsequently, Ikanovic and Mollgaard demonstrated that a substantial portion of this 93% ceiling stems from temporal stationary states rather than true mobility trajectory predictability [5]. The entropy formulation of Song et al. addresses the query “*where will the user be during the next time interval*”, which defaults to the user’s current position during stationary periods. By reformulating the target to non-stationary transition events (“*where will the user move next*”, with stationary repetitions removed), the theoretical ceiling drops to 71%; the dashed

section of the second bar is the share lost to standing still. Approximately one quarter of apparent trajectory predictability is therefore attributable to stationary temporal residence.

The two lower bars show what this correction means in practice. On a raw trajectory file, the persistence baseline alone achieves 69.2%, leaving virtually no margin below the 71% ceiling. On active mobile subscribers who genuinely move, the same rule drops to 50.2% without any change to its own definition (Chapter 3), and the fine-tuned model reaches 54.5%. A raw day file and a filtered cohort of active subscribers therefore cannot be compared on absolute accuracy: rigorous model evaluation requires measuring performance gains relative to the persistence baseline rather than reporting uncalibrated absolute accuracy figures.

Two additional insights from Song et al. directly inform subsequent sections of this report [13]:

Constancy of Predictability Ceilings Across Travel Radii. The predictability ceiling remains largely invariant to user travel distances beyond a radius of ten kilometers. However, localized predictability derived from individual visit frequencies varies widely across subscriber cohorts, peaking near 30%. Consequently, aggregate population averages mask underlying subgroup variations, necessitating the individual user profiling detailed in Chapters 5 and 7.

Diurnal Regularity Patterns. By partitioning weekly trajectories into 168 hourly slots, Song et al. observed that individual regularity reaches approximately 90% during night hours, while exhibiting pronounced dips around midday and early evening (18h–20h) [13]. This empirical distribution of temporal difficulty motivates the hourly feature integration evaluated in Chapter 6.

2.2 Categorization of Literature Contributions

The secondary literature informs the evaluation framework and model architecture utilized in this research. Luca et al. present a comprehensive survey of deep learning approaches for human mobility modeling [8]. Two standard conventions from their survey are adopted here: integrating explicit temporal representations (hour of day) into the model input, and evaluating prediction accuracy across multiple rank thresholds (top-1, top-3, and top-5). Parallel work by Qin et al. adopts identical multi-depth evaluation metrics [11]. Top- k evaluation reflects practical deployment requirements, where network operators benefit from pre-allocating handover capacity across candidate top-ranked cell targets. Furthermore, mobility surveys highlight the importance of recurrent memory mechanisms to capture historical trajectory patterns, a capability inherently supported by Transformer-based attention mechanisms.

Recent developments in LLM-based mobility prediction diverge into two principal paradigms. Wang et al. utilize zero-shot and few-shot natural language prompting without modifying underlying model parameters [14], an approach evaluated in parallel by team members Tiphain and Benjamin. Conversely, Qin et al. update model parameters directly using parameter-efficient fine-tuning across mobility datasets [11], establishing the methodological foundation adopted in this report.

The theoretical framework for user cohort stratification in Chapter 7 originates from demographic mobility studies. Pappalardo et al. demonstrate that human populations partition into distinct mobility behavioral classes rather than conforming to a single continuous spec-

trum [9]. Jeong et al. provide the algorithmic motivation: because individual historical traces are frequently sparse, grouping subscribers with similar mobility patterns enables collective parameter sharing and collaborative prediction [6]. The four user profiles established in Chapter 7 extend this concept by stratifying users according to temporal cell transition distributions.

2.3 Comparison Between Antenna-Level and Location-Level Prediction

The majority of existing literature focuses on predicting semantic *locations* (e.g., home, workplace, or clustered points of interest). In contrast, the objective of this study is the prediction of specific cellular *antennas*, presenting distinct domain-specific properties. Cellular antenna prediction is primarily addressed within mobile network engineering; for instance, Chen et al. predict physical antenna handovers using radio signal measurement logs to optimize network handover management [1].

Predicting cellular antennas strictly from sequence traces without physical signal strength telemetry introduces specific challenges. In the evaluation dataset, 12.4% of consecutive records exhibit cell identifier changes caused by inter-antenna handovers on the same base station mast without physical subscriber displacement (Chapter 3). Consequently, model evaluation involves predicting network handover decisions alongside physical human mobility. Furthermore, grading accuracy at the granular cell level represents a finer-grained target than spatial region clustering evaluated in theoretical predictability bounds.

3 Dataset Analysis and Preprocessing

3.1 Data Analysis Objectives

Prior to model training, the structural characteristics of the mobile network logs must be thoroughly analyzed. Two fundamental properties constrain all subsequent experiments: a network log event does not necessarily imply spatial displacement, and an individual cell identifier does not uniquely define a geographical site. These structural features account for several non-intuitive empirical findings presented in later chapters.

3.2 Structure of Call Detail Records

The empirical evaluation uses fifteen daily log files spanning 12 to 26 March 2014 from the Karlovy Vary study area in north-west Bohemia. The dataset consists of comma-separated values (CSV) files, where **each line represents a single subscriber's 24-hour activity trace** from 00:00:00 to 23:59:59. Depending on the specific day, daily files contain between 80,217 and 99,504 subscriber trajectories; the primary benchmark dataset from 12 March 2014 comprises 85,946 subscribers.

Each record row begins with eight metadata columns followed by a sequence of timestamped cell interaction pairs. The initial columns indicate an anonymized subscriber ID, age, gender, an administrative code, a behavioral category, a designated home cell ID, an activity cell ID, and the total event count. Following the metadata header, trajectory entries consist of **a cell identifier string paired with an event timestamp** expressed in elapsed seconds since midnight ($0 \leq t \leq 86,399$). Model training exclusively utilizes the subscriber ID, total event count, and timestamped cell identifier sequence; remaining metadata fields contain recorded anomalies (such as invalid age values) and were excluded.

Network log events are registered whenever a mobile terminal interacts with a base station mast, including automatic periodic network polling. Base stations initiate periodic location updates on fixed time intervals (typically every 4 hours on 2G networks and every 2 hours on 3G networks) to track active devices for incoming call routing. Consequently, **a network record does not imply physical movement**: a stationary subscriber generates periodic events throughout the day. The number of generated logs depends on the radio network generation as well as physical subscriber behavior. Contiguous sequences of identical cell identifiers account for the high baseline performance of the persistence rule.

3.2.1 Baseline Performance and Population Dependencies

The **persistence baseline** evaluates the prediction heuristic stating that the next cell identifier matches the most recent observed cell identifier. This deterministic baseline is evaluated across identical trajectory points and evaluation protocols as the fine-tuned language models.

Crucially, the accuracy of the persistence baseline is not fixed; it is a parameter of the evaluated population cohort. On unfiltered daily logs containing a high proportion of stationary subscribers, the persistence baseline achieves 69.2% accuracy. When the dataset is filtered to include only active mobile subscribers, baseline performance drops to 45.5%. The

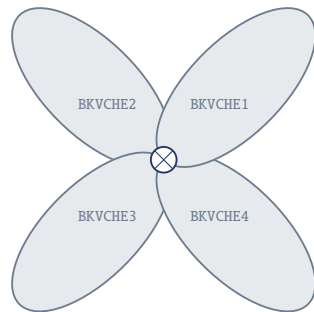
underlying baseline heuristic remains unchanged while the population mobility characteristics shift.

3.3 Distinction Between Physical Masts and Logical Cells

Cell identifiers such as BKVCHE1 represent logical network coverage sectors rather than physical site coordinates. In technical documentation, this string is designated as the **cell_id**.

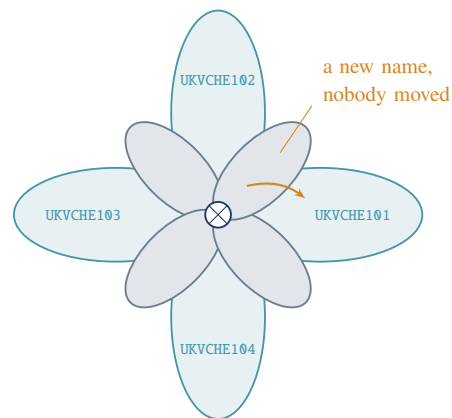
Figure 3.1 illustrates the spatial mapping of a single physical base station mast to its associated cell identifiers, viewed from above. The central crossed circle represents the physical installation, while each shaded lobe corresponds to the spatial coverage area of a directional antenna. A single physical site operates multiple directional antennas operating across different radio frequencies and sectors. **Figure 3.1.a** illustrates a 2G configuration operating four directional sector cells, while **Figure 3.1.b** adds four 3G directional cells with distinct frequency ranges and spatial orientations. Because all eight coverage lobes originate from a single physical mast location, a stationary device at the foot of the mast sits inside multiple overlapping coverage sectors simultaneously. Radio signal fluctuations can trigger network handovers between adjacent directional sectors or frequency bands, which register in raw network logs as cell transitions despite zero subscriber spatial movement. **Figure 3.2** measures how often this happens.

3.1.a One mast carries four 2G cells



one installation, four names in the data

3.1.b The same mast also carries 3G: eight cells



the smaller lobes are the 2G cells of 3.1.a

Eight names, one place. A phone lying still at the foot of the mast sits inside several lobes at once, and the network can hand it from one to the next for reasons of signal alone. Every such handover is written into the data as a change of antenna, and looks exactly like a movement.

Figure 3.1: One mast, and all the names it answers to

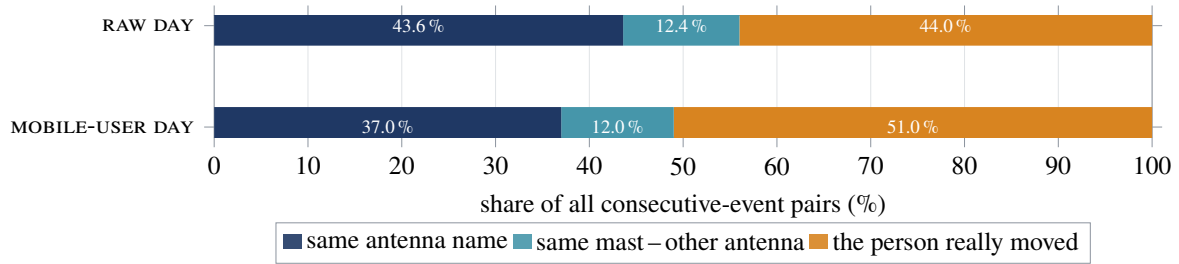


Figure 3.2: The three things a pair of back-to-back records can be

Figure 3.2 takes every pair of consecutive records in a file; each bar is 100% of one file’s pairs, split into three structural categories:

1. **Same Cell ID** (dark blue): consecutive events log identical cell identifiers, indicating stationary temporal residence.
2. **Same Mast, Different Cell ID** (teal): consecutive events log different cell identifiers mounted on the same physical base station mast, representing radio sector handovers without spatial displacement.
3. **Different Mast** (orange): consecutive events log a transition to a different base station site, representing true physical subscriber mobility.

On an unfiltered raw day file (the `RAW DAY`), only 44.0% of consecutive event pairs represent physical spatial displacement, while **12.4 out of every 100** are same-mast sector handovers that look like movement in the raw data and are not. Filtering the corpus to active mobile subscribers (the `MOBILE-USER DAY`) increases the proportion of true physical movement to 51.0%, while intra-site sector handovers remain constant at 12.0%. This constancy confirms that intra-site sector handovers are an intrinsic property of the radio cellular infrastructure rather than individual subscriber behavior. To accurately evaluate physical mobility, displacement analyses in this report assess movement at the base station mast level, while the model prediction target remains the full antenna `cell_id` string; the cost of that choice is examined in Chapter 5.

3.4 Data Preprocessing and Dataset Construction

Figure 3.3 details the four-stage data preprocessing pipeline that transforms one raw daily record, as described above, into formatted training sequences:

1. **Trajectory Extraction:** administrative header metadata columns are stripped, retaining only cell attachment sequences and event timestamps.
2. **Temporal Token Prepending:** event timestamps are converted into explicit hourly tokens (`<H00>`–`<H23>`) prepended to each cell identifier.
3. **Subscriber Sequence Assembly:** single-subscriber daily attachment records are formatted into ordered sequence strings.
4. **Stratified Population Split:** subscribers are partitioned into disjoint training and held-out test cohorts prior to model training.

As depicted by the separate training and test circles, partitioning is performed strictly by subscriber ID, ensuring no individual appears in both splits and preventing memorization leakage. Trajectory volume stratification ensures both cohorts maintain identical distributions

of short and long daily histories.

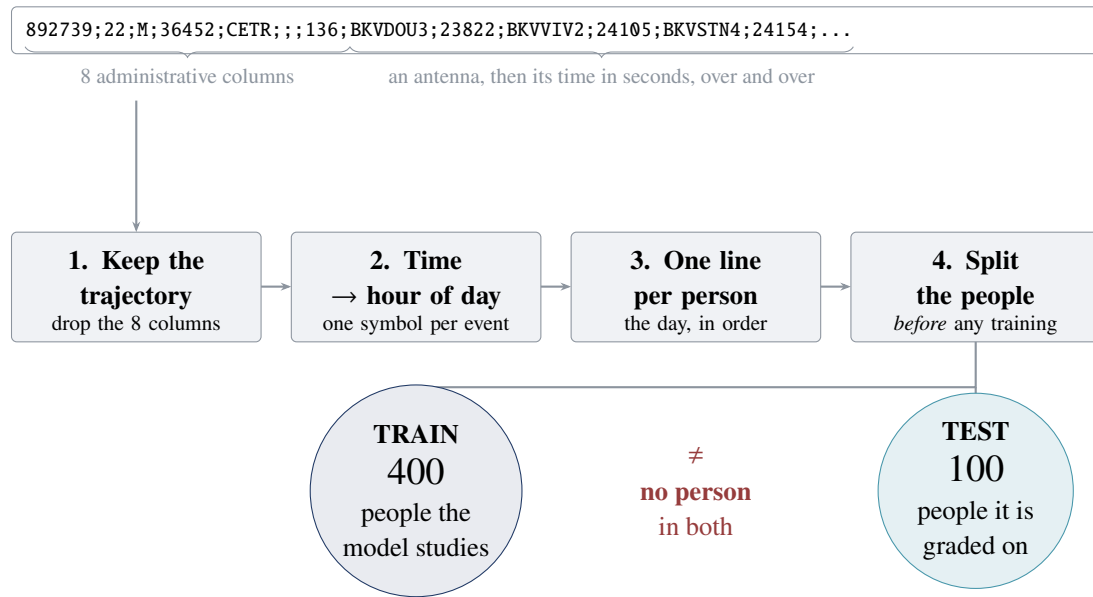


Figure 3.3: From a raw file to training material, in four steps

3.4.1 Characteristics of Experimental Cohorts

Table 3.1 summarizes the six experimental dataset cohorts created during this study: training and evaluation subscriber counts (always distinct subscribers), average daily record volume per subscriber, and the persistence baseline accuracy on the test subscribers, which is the reference every result of the corresponding chapter is compared against. The creation of distinct dataset cohorts resulted from evolving research requirements across experimental iterations: incorporating hourly features required regenerating token sequences with temporal tags; evaluating population scaling required larger single-day extractions; and multi-day scaling required concatenating eleven distinct daily logs.

Table 3.1: The six sets of people, and the score of the one-line rule on each

	Who is in it	Train	Test	Records/person	Rule
RAW DAY	one day file taken as it comes (12 March 2014)	400	100	mean ≈ 62	69.2 %
FILTERED DAY	the same day, filtered: more than 10 records and no 4-hour hole	400	100	>10 by design	79.0 %
MOBILE-USER DAY	people who move, and who stay in the area, on one day	400	100	19–200, mean 86.0	45.5 / 43.1 %
LEVELLED CORPUS	the same people, every day stretched to exactly 200 records	400	100	200 exactly	76.5 %
2,000-USER DAY	a day file sampled down to 2,000 people	900 / 1,900	100	12–200, mean 77.1	48.2 %
ELEVEN-DAY SET	11 day files of 2,000 people, stacked end to end	21,000	1,000	10–200	50.2 %

The baseline column, ranging from 69.2% to 43.1%, illustrates the population dependency established above: the rule is identical in every row, only the subscribers change. Conse-

quently, raw accuracy scores from different dataset cohorts must never be compared directly; all performance gains must be evaluated relative to the baseline of their respective cohort. For the MOBILE-USER DAY, the table reports two baseline scores: 45.5% evaluated on the final 20% of each test day, and 43.1% evaluated sequentially across all trajectory positions (a strictly harder benchmark introduced in the sixth iteration).

3.5 Scope and Methodological Limitations

Temporal Boundary: Intra-Day Predictions. The primary task evaluated in this report is intra-day trajectory prediction: forecasting the remaining segment of a subscriber’s daily trajectory given its initial prefix. Inter-day forecasting across consecutive days represents a distinct problem scope.

Spatial Coverage and Antenna Densities. The geographical study region encompasses 369 cells across 113 base stations, with individual subscribers visiting an average of twenty unique cells. At small training sizes (400 subscribers), 98.9% of cell IDs in the test set were present in the training set; at training sizes of 5,000 subscribers and above, vocabulary coverage reaches 100%. Consequently, out-of-vocabulary cell generation is not a limiting factor for larger datasets, meaning performance gains must derive from modeling transition probabilities between known cells.

3.6 Conclusion

Two structural properties of cell detail records dictate the empirical results presented in this report. First, a network log event does not necessarily signify physical spatial displacement; periodic location updates produce repeated entries during stationary periods, establishing a high persistence baseline score. Second, cell identifiers represent sectorized antennas rather than unique physical locations, with 12.4% of cell transitions occurring across sectors of the same physical mast (**Figure 3.2**). Consequently, physical mobility analyses in this report evaluate base station site transitions, while the predictive model target maintains granular cell ID resolution.

Filtering the dataset to active mobile subscribers removed trivial stationary trajectories and established a meaningful evaluation benchmark. Although late data filtering necessitated the six experimental dataset cohorts documented in **Table 3.1**, reporting performance gains relative to the persistence baseline ensures consistent comparative analysis across all experimental phases.

4 Model Architecture and Evaluation Methodology

4.1 Experimental Methodology and Objectives

Establishing a rigorous modeling and evaluation methodology required resolving two primary technical questions:

1. Given constrained computational hardware (a single GPU instance), which model architecture and parameter-efficient fine-tuning method offer feasibility and training efficiency?
2. What evaluation protocols are necessary to ensure that reported accuracy improvements represent genuine trajectory modeling rather than dataset artifacts?

The selection of model architecture and fine-tuning mechanics remained fixed across experimental iterations, ensuring consistent cross-experiment comparisons. The second question led to three fundamental evaluation principles designed to prevent false positives and over-estimation of baseline performance gains.

Data Provenance. Empirical measurements presented in this chapter derive from three early dataset cohorts documented in **Table 3.1**: raw logs (the RAW DAY), basic filtered logs (the FILTERED DAY), and active mobility logs (the MOBILE-USER DAY). Each cohort evaluates models trained on **400 subscribers** and graded on **100 held-out subscribers**. Corresponding persistence baseline accuracies across these three datasets are 69.2%, 79.0%, and 45.5%, respectively.

4.2 Parameter-Efficient Fine-Tuning and Model Architecture

Fine-tuning pre-trained language models can be conducted under varying memory footprints. Fine-tuning the full parameter set requires substantial memory resources, whereas Parameter-Efficient Fine-Tuning (PEFT) methods freeze pre-trained parameters and train auxiliary low-rank matrices.

Figure 4.1 compares three parameter adaptation mechanisms and their memory cost. Full fine-tuning (**Figure 4.1.a**) modifies 100% of model parameters, requiring memory to maintain optimizer states for all weights and exceeding the VRAM capacity of standard single-GPU instances. Low-Rank Adaptation, LoRA (**Figure 4.1.b**) [4], freezes the original base weight matrices W and inserts auxiliary low-rank decomposition matrices A and B , training only $\approx 2\%$ of total parameters. Quantized Low-Rank Adaptation, QLoRA (**Figure 4.1.c**) [2], further compresses frozen base parameters into 4-bit NormalFloat precision (reducing base memory footprint to $\approx 1/4$) while training 16-bit LoRA adapter weights. QLoRA was selected to enable model execution within 16 GB GPU memory limits. This selection primarily impacts training throughput rather than model accuracy, accelerating execution from 5.5 hours to 30 minutes per training run and supporting extensive hyperparameter sweeps and per-group adapter routing experiments (Chapter 7).

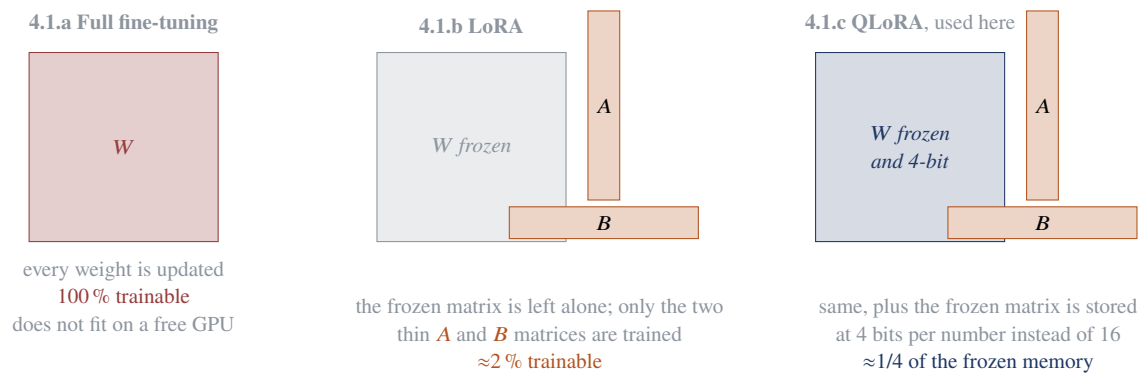


Figure 4.1: Three ways to adapt a model, and their cost in memory

4.2.1 Model Selection and Capacity Evaluation

Initial experiments evaluated **Mistral-7B** [7] (7 billion parameters). A preliminary 30-minute test run on 1.2% of the dataset yielded 16.87% top-1 accuracy, verifying pipeline end-to-end functionality. (Comparative tests using Llama models failed to produce syntactically valid cell identifier output.)

However, full training of Mistral-7B required 5.5 hours per run on a single T4 GPU, reaching a maximum accuracy of 41.1%. In contrast, **Qwen2.5-0.5B-Instruct** [12] (500 million parameters, 14× smaller) completed 49 training epochs in 72 minutes (reducing to 29 minutes under optimized early stopping). Consequently, Qwen2.5-0.5B was chosen for all subsequent experiments.

Because next-cell prediction across 369 discrete vocabulary items requires modeling localized spatial transition patterns rather than general world knowledge or natural language reasoning, the smaller model offers adequate capacity. The 14× parameter reduction provided a 23× speedup per run, enabling extensive experimental iteration within computational constraints.

Figure 4.2 illustrates the software and hardware stack utilized throughout this study, read from the bottom up:

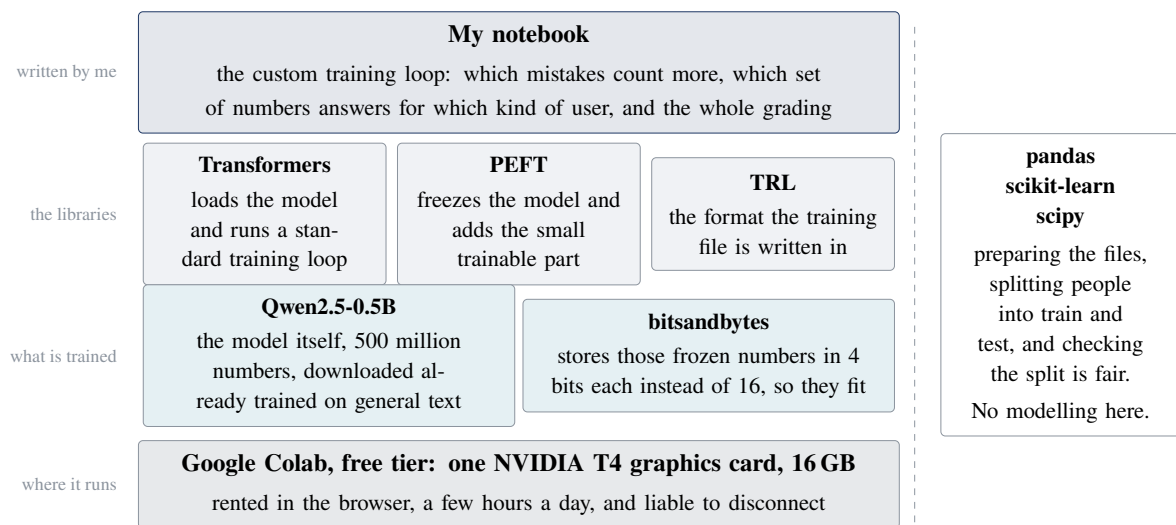


Figure 4.2: The tools, and the job of each one

- **Hardware Execution Layer** (bottom): Google Colab single NVIDIA T4 GPU instances (16 GB VRAM), establishing the primary computational memory constraint.
- **Model & Quantization Layer**: Qwen2.5-0.5B base parameters quantized to 4-bit precision using bitsandbytes.
- **Open-Source Library Layer**: Hugging Face Transformers for model loading and optimization loops; PEFT for managing LoRA adapter parameters; and TRL for sequence dataset formatting.
- **Custom Modeling Layer** (top): custom Python training and evaluation routines implementing weighted cross-entropy loss, dynamic sequence sampling, per-group LoRA adapter routing, and evaluation protocols. Reaching inside the training step to implement this loss weighting and parameter routing was the primary requirement governing the choice of the whole toolchain.
- **Analytical Tooling** (right): the standard scientific libraries pandas, scikit-learn and scipy, which handle offline data splitting, feature engineering, and statistical bootstrap resampling [3] without touching neural network parameters.

4.3 Data Representation and Tokenization Scheme

Language models process discrete token sequences. **Figure 4.3** follows one subscriber day from its storage on disk to the sequence handed to the model.

4.3.a One person’s day, as it is stored on disk. One line of the training file, spread over several lines here so that it can be read. Nothing is a conversation in any real sense: the three labels are simply the slots the training library expects to find.

```
{
  "conversations": [
    {
      "role": "system",
      "content": "You are an AI that generates a user's cell-visit trajectory."
    },
    {
      "role": "user",
      "content": "Here is the sequence for this user."
    },
    {
      "role": "assistant",
      "content": "User 2832452 | 200 events | BSOCHO1 BSOCHO1 UKVCHE102 BSOCHO1 ..."
    }
  ]
}
```

system	What the task is. The same words on every line of the file.
user	The request. Also the same everywhere: a placeholder, not a question.
assistant	The person’s actual day. This is what the model has to learn to produce, and the only part its mistakes are counted on.

4.3.b The same line, once it has been cut into symbols. This is what the model actually reads. Each antenna name is *one* symbol, not a spelling; each event is preceded by its hour. Only the dark blue positions are graded.

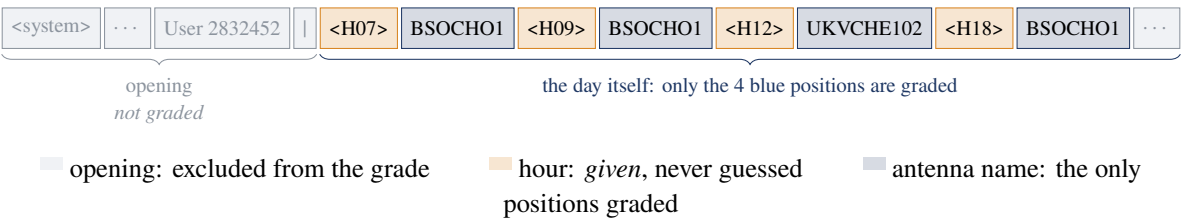


Figure 4.3: One person’s day, as the model receives it

On disk (**Figure 4.3.a**), each day is a JSON Lines record with system, user and assistant role slots, a layout imposed by the TRL library: the first two slots are boilerplate and the trajectory sits in the third. Once tokenized (**Figure 4.3.b**), the day becomes a sequence built on three encoding choices:

1. **Dedicated Vocabulary Tokens for Cell IDs:** all 369 cell identifiers were added to the model’s vocabulary as atomic tokens (dark blue). Subword tokenization can no longer split a string like BSOCH01 into pieces, so the model chooses among 369 known answers instead of spelling one out.
2. **Explicit Hourly Tokens:** an hour token (<H00> through <H23>, orange) precedes each cell event. It is an input given for free, never predicted, and costs one token position per event instead of about five for a text string like “07:31”, which would have cut the visible history by roughly two thirds.
3. **Loss Masking:** cross-entropy loss is computed exclusively on cell identifier positions (dark blue); the header boilerplate (grey) and the hour tokens are excluded from the grade.

4.3.1 Training Pipeline and Optimization Mechanics

Figure 4.4 details the five-stage training loop:

1. **Sequence Slicing:** sampling sub-sequence windows of timestamped cell events.
2. **Target Masking & Forward Pass:** predicting the next cell token given the preceding context.
3. **Surprise Loss Computation:** calculating the cross-entropy loss between the predicted probabilities and the true target cells.
4. **Parameter Gradient Updates:** backpropagating gradients to the trainable LoRA adapter weights only ($\approx 2\%$ of parameters), the base parameters remaining frozen.
5. **Validation Checkpointing & Early Stopping:** evaluating accuracy on held-out validation subscribers after each training pass and saving the best checkpoint.

From the sixth experimental iteration onward, two of these steps were modified. At Step 3, **weighted cross-entropy loss** scales the loss at specific hourly positions by a multiplier (e.g. $4\times$ for 18h–20h transitions), emphasizing transition errors without duplicating data. At Step 1, **targeted sequence sampling** draws training slices containing key transition windows up to $3\times$ more frequently during batch generation.

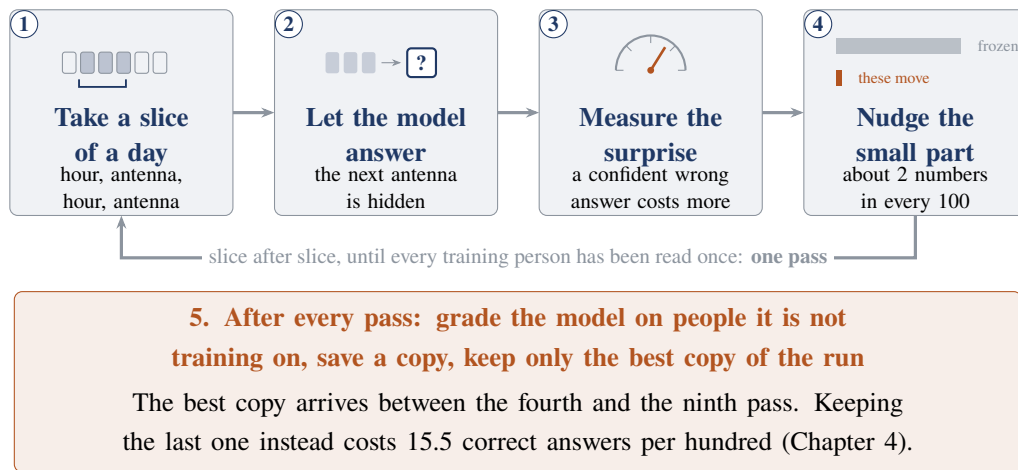


Figure 4.4: The five steps of one training run

4.3.2 Hardware Environment and Computational Resources

All reported models were trained on **Google Colab** instances with a single NVIDIA **T4 GPU** (16 GB VRAM), the bottom layer of **Figure 4.2**; no other hardware was used.

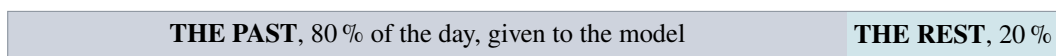
4.3.3 Integration of Peer Contributions

Data extraction tooling for generating daily trajectory logs and active mobility cohorts was developed by Nino. The framing of active mobility trajectory filtering was developed by Tiphain and Benjamin. Trajectory length equalisation techniques evaluated in Chapter 8 adapt methods formulated by Tiphain.

4.4 Evaluation Protocol and Metrics

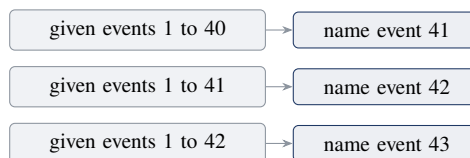
Model evaluation follows the standardized protocol of **Figure 4.5**.

4.5.a Each person's day is cut in two, once and for all



where this cut falls is called the *context fraction*; several experiments move it from 30 % to 90 %

4.5.b The rest is then answered one moment at a time, never in one go



each time the real past, never the model's own earlier answers

Figure 4.5: The cut of the day, and the answering of each moment

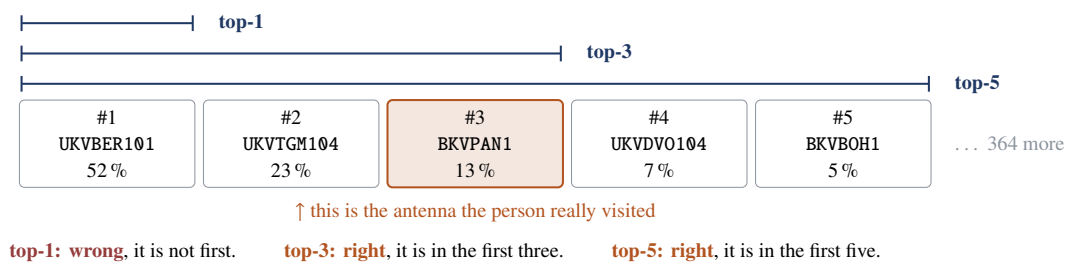
Each test trajectory is split once into a historical prefix handed to the model and a target suffix to be predicted, by default 80% and 20% of the day (**Figure 4.5.a**). Where that cut falls is part of the measurement, not a property of the model (Principle 2 below). The suffix is

then evaluated step by step (**Figure 4.5.b**): the model is asked for the single next cell, graded against ground truth, and given the true event before the next step. It therefore never sees the future and never compounds its own errors, and the persistence baseline is run on exactly the same steps with exactly the same history, which is what makes the two scores comparable.

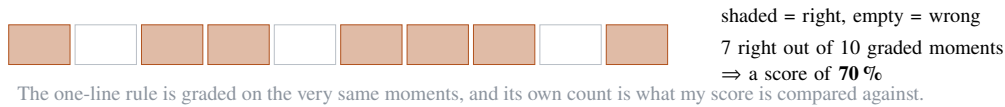
Figure 4.6 shows how each step is graded. The model does not return a single answer but a probability distribution over all 369 candidate cell tokens, sorted from most to least likely (**Figure 4.6.a**). **Top-1** accuracy asks whether the true cell ranks first; **top-3** and **top-5** whether it appears among the first three or five candidates. In the example drawn, the true cell ranks third, so the same prediction fails at top-1 and succeeds at the other two depths. Reporting these three depths follows the convention of the mobility literature [8, 11]. Accuracy itself is a plain count (**Figure 4.6.b**): correct steps divided by evaluated steps across the held-out test subscribers, ten steps with seven correct giving 70%. The persistence baseline is counted on the same steps, and every result in this report is the difference between the two counts.

4.6.a The model never answers once. It ranks all 369 antennas.

Here are its first five, for one moment of one person's day, with how sure it is of each.



4.6.b The score is then just a count, over every graded moment of every test person.



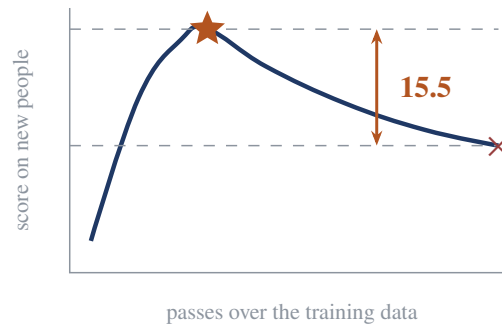
The one-line rule is graded on the very same moments, and its own count is what my score is compared against.

Figure 4.6: Top-1, top-3 and top-5, and the way a score is obtained

4.5 Evaluation Methodology Principles

4.5.1 Principle 1: Early Stopping and Overfitting Mitigation

This principle corresponds to **Figure 4.7**. Fine-tuning language models on small trajectory datasets leads to severe overfitting when trained for too many epochs. **Figure 4.8** shows four training runs on four datasets: the horizontal axis counts training passes, the vertical axis is accuracy on unseen held-out subscribers, the solid line is the model with a star at its best pass, and the dashed line is the persistence baseline on the same subscribers. In all four panels, validation accuracy peaks within 4 to 9 passes and then declines, while training loss keeps decreasing: the model has moved from learning how the population moves to memorizing its training examples.



The score on unseen people peaks early, then decays for tens of passes. Reading the **end** of the run instead of its **best point** throws away 15.5 correct answers per hundred.

Figure 4.7: Never read the last pass of a run

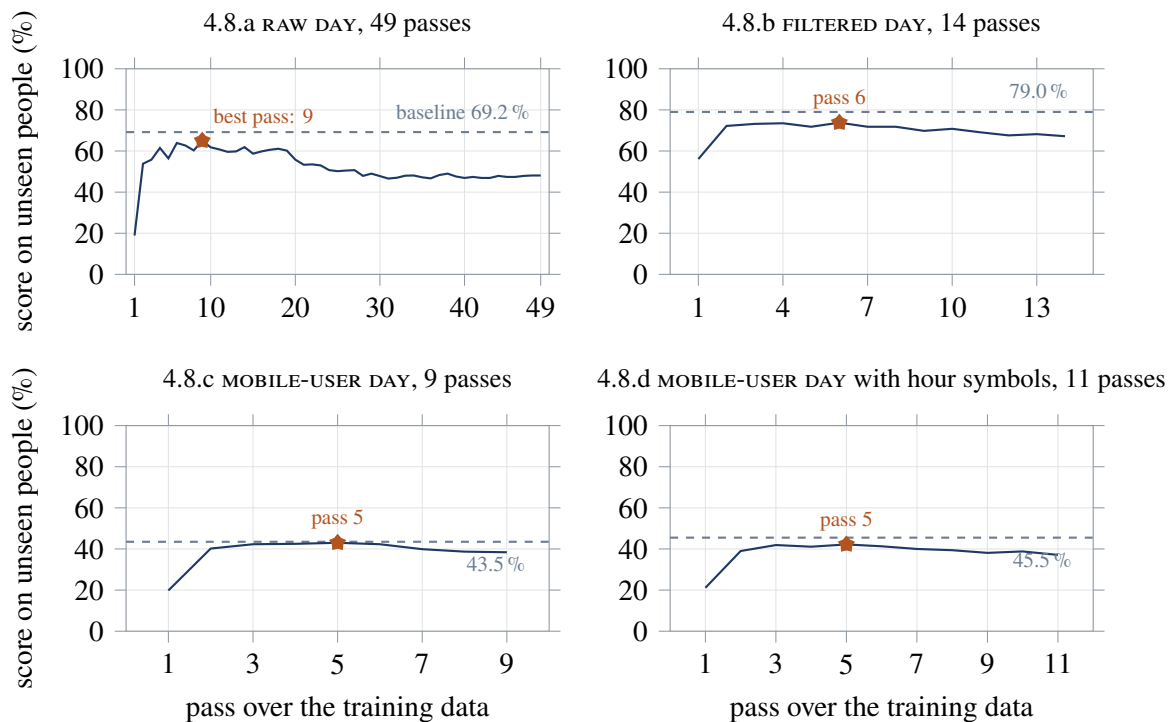


Figure 4.8: The score on unseen people, pass by pass, on four sets of people

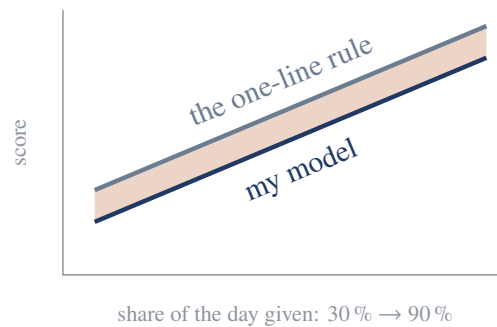
→ Produced by `train_cellid_llm.ipynb`, the training loop these per-pass scores come from.

In the extended 49-epoch run of **Figure 4.8.a**, validation accuracy peaked at epoch 9 (65.0% validation, 64.2% test) before degrading to 48.1% (48.7% test) at epoch 49. Reading the final epoch instead of the peak would cost **15.5 %** of test accuracy on held-out subscribers. Early stopping at peak validation accuracy, alongside regularization (dropout 0.1, weight decay, learning rate scheduling), preserved generalization and reduced training time from 72 to 29 minutes. The dashed baseline moves from panel to panel (69.2% on the unfiltered day of **Figure 4.8.a**, 43.5% once only mobile subscribers are kept) because the subscribers differ, so the panels are comparable only on the gap between solid and dashed line; in all four, that gap is negative. All subsequently reported metrics evaluate the checkpoint saved at peak

validation accuracy.

4.5.2 Principle 2: Context Fraction and Evaluation Split Sensitivity

This principle corresponds to **Figure 4.9**. The prefix context fraction (share of the trajectory provided to the model before prediction) directly affects task difficulty and baseline performance. **Table 4.1** presents model performance across context fractions from 30% to 90% on the **FILTERED DAY** and the **MOBILE-USER DAY**: **Rule** is the persistence baseline on the same steps, and **Distance**, model minus baseline, is the only column that compares like with like. Moving the cut alters the baseline score itself, because it shifts which event counts as “the most recent cell.” On the **FILTERED DAY**, evaluating at an 80% context fraction yields 80.1% model accuracy against a 79.0% persistence baseline (+1.1%), whereas evaluating at a 70% context fraction trails the baseline by 0.3%. On the **MOBILE-USER DAY**, absolute model accuracy increases from 41.5% to 47.6% as the context fraction expands from 30% to 90%; however, baseline accuracy increases at an identical rate (42.9% to 50.9%), leaving the performance delta negative across all cuts (these the **MOBILE-USER DAY** rows are one model read at seven cuts, hence their identical **Valid.** column, whereas each the **FILTERED DAY** row is a separate run). Consequently, apparent accuracy gains resulting strictly from expanding historical context represent measurement ruler artifacts rather than model improvements, requiring context fractions to be held constant during comparative evaluations.



Giving the model more history raises its score, and raises the rule by the same amount, because it also moves what counts as “the last antenna seen”. A rising score across cuts says nothing about the model.

Figure 4.9: The cut of the day belongs to the ruler

Table 4.1: Moving the cut moves the ruler with it

past / rest	Valid.	Model	Rule	Distance	top-3	top-5	Beats it?
<i>FILTERED DAY</i> , three separate trainings, one per cut							
70 / 30	72.3 %	76.5 %	76.8 %	−0.3	87.1 %	90.1 %	no
80 / 20	73.8 %	80.1 %	79.0 %	+1.1	89.1 %	91.6 %	yes
90 / 10	75.7 %	77.7 %	80.1 %	−2.4	87.2 %	90.5 %	no
<i>MOBILE-USER DAY</i> , one training at 80/20, evaluated at seven cuts							
30 / 70	43.0 %	41.5 %	42.9 %	−0.9	62.1 %	69.9 %	no
40 / 60	43.0 %	42.0 %	43.2 %	−0.6	62.1 %	69.7 %	no
50 / 50	43.0 %	42.0 %	43.3 %	−0.7	62.0 %	69.4 %	no
60 / 40	43.0 %	42.7 %	44.0 %	−0.9	62.1 %	69.5 %	no
70 / 30	43.0 %	43.6 %	44.3 %	−0.3	62.1 %	69.5 %	no
80 / 20	43.0 %	44.6 %	45.5 %	−0.9	63.9 %	70.7 %	no
90 / 10	43.0 %	47.6 %	50.9 %	−2.1	68.0 %	74.3 %	no

↪ Produced by `train_cellid_llm.ipynb`, the context-fraction sweeps.

4.5.3 Principle 3: Baseline Upper Bounds and Task Room Constraints

This principle corresponds to **Figure 4.10**. On unfiltered daily logs, the persistence baseline achieves 69.2% accuracy due to high stationary periods. A model achieving 64.4% accuracy on such raw data has not failed in learning mobility patterns; rather, it has been evaluated on a benchmark dominated by stationary trivial predictions. Restricting benchmarks to active mobile trajectories (reducing baseline score to 45.5%) creates a viable dynamic evaluation range.



Blue is what one line of code already takes; the pale part above it is what a model could still win. Choosing whom to measure on nearly doubled that room, from 30.8 to 54.5.

Figure 4.10: A test the one-line rule already passes

4.5.4 Methodological Implications for Hyperparameter Optimization

Applying these three evaluation principles summarized in **Figures 4.7, 4.9 and 4.10** revealed that hyperparameter tuning alone yields limited performance improvements on uniform co-

horts. Individual predictions revealed stark disparities: stationary trajectories were predicted with near 100% confidence, whereas active transition trajectories resulted in incorrect baseline-like predictions. This variance across individual subscriber profiles motivated the individual user analysis presented in Chapter 5.

4.6 Conclusion

This chapter established a standardized model fine-tuning architecture (Qwen2.5-0.5B fine-tuned via QLoRA with explicit cell and hour tokenization) and a rigorous evaluation protocol. Model capacity and PEFT mechanics remained fixed, allowing subsequent experimental gains to be attributed strictly to data representation and user conditioning mechanisms.

Establishing early stopping (**Figure 4.8**), standardized context fraction splits (**Table 4.1**), and active mobility cohort benchmarking eliminated false positive findings and provided a reliable foundation for subsequent research chapters.

In hindsight, these three principles weigh more than any setting tuned in this chapter. Reading the best saved copy instead of the last pass is worth 15.5 correct answers per hundred on its own (**Figure 4.7**), far more than any later gain of the project (at most +4.3 %). Moving the cut of the day raised the model from 41.5% to 47.6% while the baseline rose from 42.9% to 50.9% (**Table 4.1**), so a rising score was never evidence on its own. Measuring on people who move nearly doubled the room left to a model, from 30.8 to 54.5 correct answers per hundred (**Figure 4.10**). Every result in the following chapters is read under these three rules.

5 Impact of Individual User Characteristics on Predictability

5.1 Objectives and Predictability Determinants

Identical fine-tuned models exhibit extreme performance variance across individual subscribers, yielding near 100% prediction accuracy for certain individuals while scoring near zero for others. This variance indicates that aggregate population accuracy is incomplete without individual trajectory breakdown. The objective of this chapter is to resolve the following question: *which intrinsic trajectory properties determine individual predictability?* If predictability depends primarily on hyperparameter tuning, model capacity should be increased. If predictability depends on individual subscriber mobility traits, aggregate performance is fundamentally constrained prior to training. The empirical results presented in this chapter demonstrate that individual subscriber characteristics dictate overall predictability.

Data Provenance. Analysis uses two dataset cohorts from **Table 3.1**: raw daily logs evaluated on 1,173 subscribers, and active mobility logs (the MOBILE-USER DAY) evaluating 400 training subscribers and **100 held-out subscribers**.

5.2 Correlation Analysis of Individual User Features

To identify trajectory features governing predictability, five subscriber-level candidate metrics computable directly from individual traces were evaluated: cell transition frequency, primary cell concentration, distinct cell count, transition temporal entropy, and total log event volume.

Each candidate metric was correlated with individual model prediction accuracy using the **Pearson correlation coefficient** (r), formulated according to Pearson [10]:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (5.1)$$

where n represents subscriber count, x_i denotes candidate feature value for subscriber i , y_i represents top-1 prediction accuracy for subscriber i , and $\bar{x}$, $\bar{y}$ denote sample means.

Figure 5.1 ranks the candidate metrics by their Pearson coefficient r , measured on 1,173 subscribers of the RAW DAY (dark blue) and on 100 held-out subscribers of the MOBILE-USER DAY (teal); a bar pointing left means that more of the property goes with lower accuracy. The primary determinant of predictability is cell transition frequency, formalized as the **cell-switch rate**: the proportion of consecutive event records in a subscriber trajectory that register a change in cell identifier, near 0% for stationary subscribers and approaching 100% for highly mobile ones. It correlates inversely with prediction accuracy, at $r = -0.971$ on the RAW DAY and $r = -0.773$ on the MOBILE-USER DAY: *higher cell transition frequency goes with lower prediction accuracy*. The first value is close to arithmetic, since a subscriber’s cell-switch

rate is the mirror image of persistence baseline accuracy on that subscriber, so any model imitating the baseline earns a strong negative coefficient before anything is learned. Primary cell concentration ($r = +0.696$) and transition temporal entropy ($r = -0.811$) reflect the same underlying repetitiveness from other angles. The remaining features stay below $|r| = 0.24$: unique cell count ($r = -0.232$), prediction instance count ($r = -0.101$) and total trajectory length ($r = +0.054$) exert negligible influence on individual predictability.

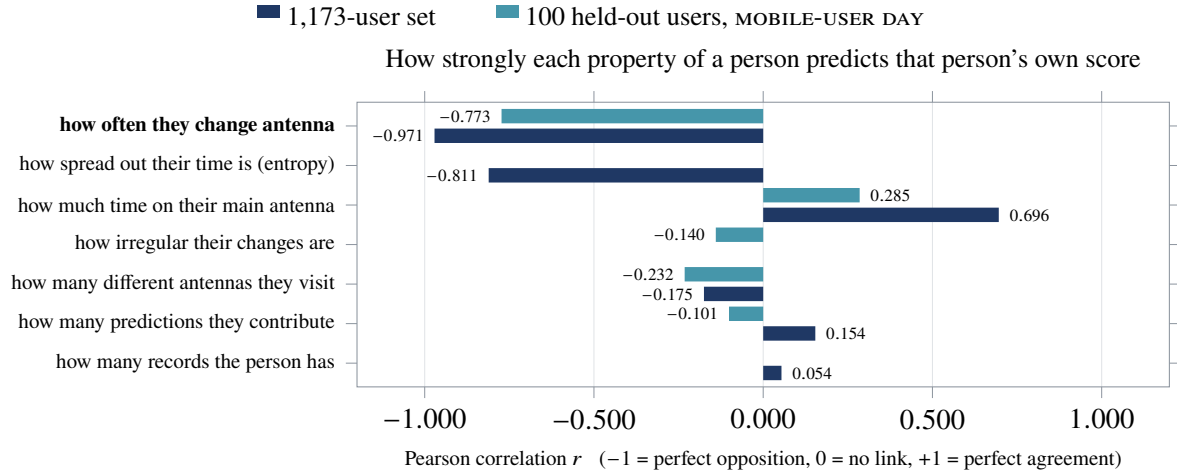


Figure 5.1: The one property that predicts a person's score

5.2.1 Quintile Stratification Analysis

Figure 5.2 tests this dependency without relying on a scalar coefficient: subscribers are sorted by cell-switch rate and cut into equal cohorts, quintiles of 500 subscribers on the RAW DAY in **Figure 5.2.a** and quartiles of 100 held-out subscribers on the MOBILE-USER DAY in **Figure 5.2.b**, with mobility increasing from left to right. In each pair of bars, grey-blue is the persistence baseline and dark blue the fine-tuned model, both scored on exactly the same moments, and the number above the pair is the difference between them.

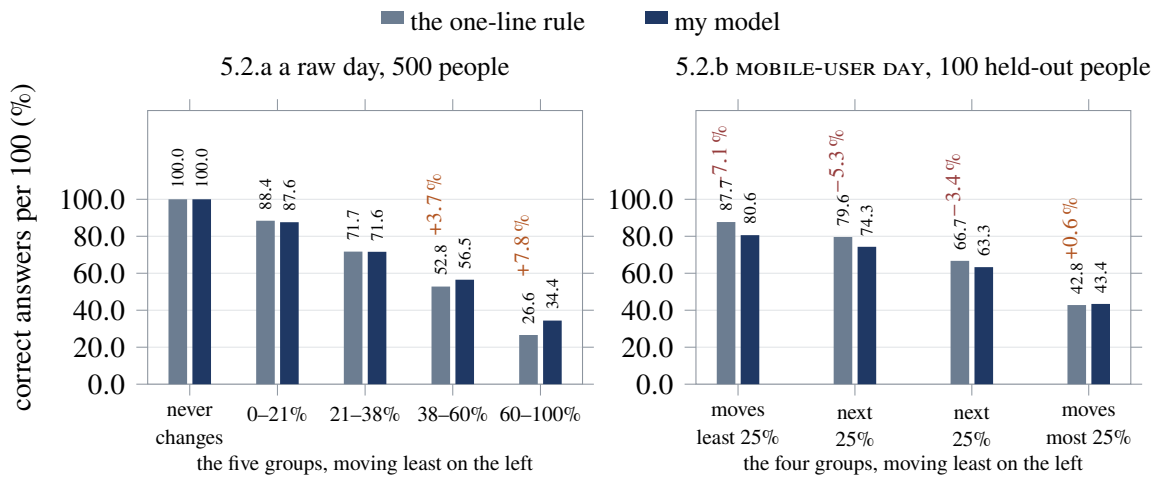


Figure 5.2: The score follows the rate at which a person changes antenna

To verify that individual difficulty reflects intrinsic subscriber behavior rather than an artifact of the evaluation prefix length, cross-context stability was evaluated: individual

accuracy at a 30% context fraction and at an 80% context fraction correlate at $r = 0.77$ (**Table 6.1** in Chapter 6), confirming that predictability is an intrinsic property of the subscriber trajectory.

5.2.2 Performance Breakdown across Restlessness Segments

On the `RAW DAY` (**Figure 5.2.a**), the lowest quintile, subscribers who never change cell ID, is predicted at 100% by both the model and the baseline, while the highest quintile plateaus at 34.4%: 65 correct answers per hundred separate the two ends of a single population under one trained model. Across the three lowest quintiles the two bars are virtually identical. The model pulls ahead only in the two highest quintiles, by +3.7 % (56.5% vs. 52.8%) and +7.8 % (34.4% vs. 26.6%), and these hold only 18.5% of the evaluated prediction instances, the highest quintile alone 5.6%. A +7.8 % improvement on 5.6% of the evaluation volume barely shifts aggregate accuracy. On the independent held-out cohort of **Figure 5.2.b**, the ordering is the same but the result is negative: the model trails the baseline in the three lower mobility quartiles (−7.1 %, −5.3 % and −3.4 %) and draws level (+0.6 %) only in the highest. This establishes a key reporting convention: *aggregate population accuracy averages distinct mobility cohorts; reporting segment-specific accuracy relative to baseline benchmarks is required to capture true model contributions.*

5.3 Impact of User Activity Volume on Aggregate Metrics

Evaluating prediction volume distribution reveals significant inequality across subscribers. Partitioning 1,173 subscribers into quintiles based on event volume demonstrates that the highest-volume decile (117 subscribers) contributes **58.6%** of the 54,173 total evaluated prediction instances. Consequently, unweighted aggregate accuracy metrics are heavily dominated by this sub-population.

However, event volume does not correlate strongly with prediction accuracy ($r = +0.154$). Higher event volume corresponds to slightly lower cell-switch rates ($r = -0.150$), explaining the minor upward accuracy trend in high-volume cohorts.

5.4 Analysis of Cell-Level vs. Mast-Level Prediction Errors

Analyzing error cases at individual subscriber resolution reveals that a portion of prediction errors stems from cell-level vs. base-station-level definitions.

Table 5.1 lists the seven lowest-scoring subscribers of the whole test set on the `FILTERED DAY`, every one at or below 12.5% accuracy. `user_id` is the anonymous number the operator assigned to the subscriber for that day, **Moments** the number of evaluated moments they contribute, and **Ceiling** the best score reachable given the cells present in their own history. These worst cases fall into three distinct failure modes:

1. **Low record volume** (rows 1–4): subscribers contributing between 2 and 12 evaluation moments, where small samples produce high variance (3 moments can only yield 0%, 33%, 67% or 100%).

2. **Genuine trajectory change** (row 5): a learnable behavioral shift that both baseline and model miss (ceiling 89%, baseline 67%, model 11.1%).
3. **Intra-site sector hand-offs** (rows 6–7): subscribers oscillating between adjacent sector antennas of the same physical mast (e.g. UKVCHE102 and UKVCHE104). The ceiling is 100% and the baseline reaches 50%–67% by repeating sector IDs, while the model, attempting to forecast the sector switches, scores 0% despite zero physical displacement.

Table 5.1: The seven worst-scoring people, and the reason each scores badly

user_id	Moments	Ceiling	Rule	Model	Which situation
1049149	3	0 %	0 %	0 %	A, two to twelve moments each: too little material to judge a model
1291964	3	0 %	0 %	0 %	
1344347	8	25 %	12 %	12.5 %	
929903	12	75 %	8 %	8.3 %	B, a real change of behaviour
1161518	9	89 %	67 %	11.1 %	
828479	2	100 %	50 %	0 %	
1067194	3	100 %	67 %	0 %	C, two names, one mast

↪ Produced by `train_cellid_llm.ipynb`, the per-user error listing.

Maintaining granular `cell_id` target prediction fulfills the cell-level forecasting mandate, even though a fraction of prediction errors represents intra-site sector handovers rather than geographical mobility.

5.5 Conclusion

Three empirical findings summarize this chapter:

1. Individual trajectory predictability is dictated primarily by cell-switch frequency (repetitiveness).
2. Fine-tuned language model accuracy gains concentrate within high mobility cohorts representing a minor fraction of total evaluation events.
3. Intra-site sector transitions account for a systematic fraction of cell-level prediction errors.

Because scaling general capacity does not alter low-mobility baseline bounds, improving high-mobility transition forecasting requires temporal conditioning signals (hour of day) to anticipate transitions.

This chapter demonstrated that trajectory predictability is primarily governed by subscriber mobility characteristics rather than model parameter scale. Cell-switch rate represents the single strongest predictor of accuracy ($r = -0.773$), whereas trajectory length and unique cell counts exhibit minimal correlation ($|r| < 0.24$).

Model accuracy gains over persistence baselines concentrate within high mobility cohorts (+7.8 % in the highest cell-switch quintile of **Figure 5.2.a**). Because high mobility subscribers generate a minority of total event volume, aggregate population accuracy shifts marginally. These findings establish the necessity of reporting cohort-stratified accuracy deltas throughout mobility prediction research.

6 Temporal Feature Integration and Hourly Loss Weighting

6.1 Hypothesis Formulation and Temporal Encoding Objectives

Prior experimental configurations provided models strictly with ordered cell identifier sequences, excluding explicit temporal timestamps. Because human mobility exhibits strong temporal periodicity (e.g., morning and evening commutes), excluding temporal context omits a key predictive feature.

The central research objective evaluated in this chapter is to investigate whether *explicit temporal encoding (hour of day)* improves transition prediction accuracy, and whether loss weighting across specific transition hours forces the model to anticipate non-stationary mobility events.

Data Provenance. Experiments evaluate active mobility dataset the `MOBILE-USER DAY` (Table 3.1), comprising 400 training subscribers and **100 held-out evaluation subscribers**. Corresponding persistence baseline accuracy is 45.5% on 80/20 evaluation splits and 43.1% across whole-trajectory evaluation.

6.1.1 Identification of Temporal Performance Degradation Patterns

The starting point of this chapter is not a new experiment but a closer reading of the previous training. The model of Chapter 4, trained on the `MOBILE-USER DAY` without any hour token and read at seven context fractions in Table 4.1, was scored again subscriber by subscriber instead of on the whole population. This individual error analysis revealed specific trajectories where model accuracy degraded sharply as context length increased, and it is this observation that leads to the finding below.

Table 6.1 scores eight representative subscribers twice with the same trained model, once under a short history prefix (30%–60% of the day) and once under a long one (80%–90%); the **Change** column is the second score minus the first. More history should help or at worst do nothing, which is what rows 1–3 show, and across the whole population the two scores correlate strongly ($r = 0.77$), so predictability is an intrinsic subscriber trait rather than an artifact of the context split. The bottom four entries (rows 5–8), however, collapse to near 0%. With an 80%–90% prefix, the moments left to predict fall in the late afternoon and evening; without explicit temporal tokens, the model keeps repeating the morning cell, with up to 78% output probability, at the exact time these subscribers begin their evening commute. Sequence context alone, without temporal encoding, fails to distinguish late-day transition probability shifts from mid-day stationary patterns.

Table 6.1: Eight people, and the four whose score collapses when given *more* history

user_id	Short past	Long past	Change	Which situation
2046755	12.1 %	10.8 %	−1.3	A , difficulty is a trait of the person: more history changes nothing
599692	16.5 %	16.7 %	+0.2	
1243093	8.3 %	15.6 %	+7.3	
1367716	60.4 %	37.5 %	−22.9	B , a real decline, still usable
401222	32.2 %	0.0 %	−32.2	C , total collapse at the end of the day: the model repeats the last antenna with up to 78 % confidence
1300439	40.7 %	0.0 %	−40.7	
1223357	33.7 %	4.2 %	−29.6	
1153529	20.0 %	3.0 %	−17.1	

↪ Produced by `train_cellid_llm.ipynb`, the per-user, per-context evaluation pass.

6.2 Temporal Feature Representation and Experimental Hypotheses

Temporal feature integration prepends dedicated hourly tokens (<H00> through <H23>) to cell event tokens within the sequence. Two distinct hypotheses were evaluated:

Hypothesis 1 (Temporal Feature Sensitivity). Providing explicit hour tokens enables the model to condition transition probabilities on diurnal temporal states (distinguishing morning vs. evening transition likelihoods).

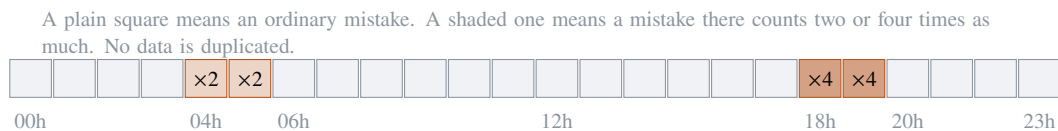
Hypothesis 2 (Targeted Loss Weighting). Assigning higher cross-entropy loss weights to specific *transition windows*¹ focuses gradient updates on high-mobility periods.

Initial loss weighting experiments assigned heuristic transition windows based on preliminary domain assumptions: **04:00–06:00** (2× loss weight) and **18:00–20:00** (4× loss weight).

6.2.1 Ablation Analysis of Temporal Tokens

To verify whether the model actively utilizes temporal tokens rather than ignoring them as sequence noise, an ablation experiment was performed.

The experiment starts from the input format drawn in **Figure 6.1**: each event enters the sequence as an hour token followed by a cell token, and errors count twice at 04h–05h and four times at 18h–19h.

**Figure 6.1:** The 24 hours of the day, and the cost of a mistake in each

The trained model is then left untouched and only its inputs are altered. **Figure 6.2** shows the three corruptions: the same model is re-evaluated on the same subscribers with the hour

¹**Transition window:** predefined temporal intervals where loss values are multiplied (e.g., 4× multiplier for 18h–20h) to penalize transition prediction errors.

tokens altered and the cell tokens left untouched.

The same trained model, the same test people, the same sequence of antennas. Only the hour symbols change.

hours left alone	<H07>	<H09>	<H12>	<H18>	the reference
hours shuffled	<H12>	<H18>	<H07>	<H09>	real hours, wrong order
all hours set to 12h	<H12>	<H12>	<H12>	<H12>	no information at all
every hour moved by +12h	<H19>	<H21>	<H00>	<H06>	order kept, labels wrong

Figure 6.2: The corruptions applied to the hours, and only to the hours

Each corruption yields one bar of **Figure 6.3**, which compares the resulting top-1 accuracies (the MOBILE-USER DAY, 80% context, 1,747 evaluated moments):

1. **Uncorrupted reference** (44.7%): correct hour tokens at every position.
2. **Constant dummy tokens** (44.0%, −0.7%): every hour token replaced by <H12>, which deletes the temporal information.
3. **Shuffled temporal tokens** (41.7%, −3.0%): hour tokens shuffled across positions, so real hours are attached to the wrong events.
4. **+12-hour label shift** (45.2%, +0.5%): every hour moved forward by 12 (<H07> → <H19>), which preserves the ordering but swaps day and night labels.

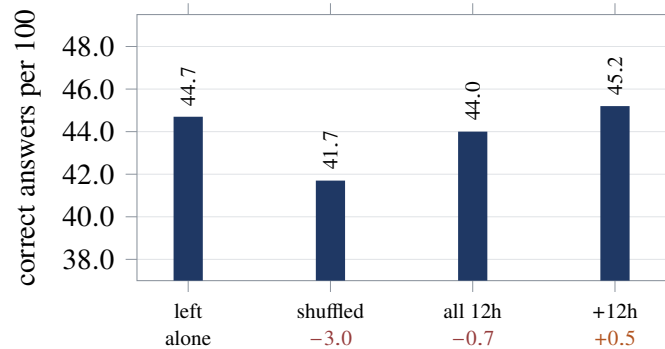


Figure 6.3: The cost of each corruption of the hours

Shuffling costs about 4× as much as deleting the hours (−3.0% vs. −0.7% in **Figure 6.3**). If the model ignored the hour tokens, corrupting their ordering could not cost more than removing them altogether.

The small overall effect of the +12h shift is not indifference either, as **Figure 6.4** shows by separating the upweighted hours from the rest: inside the upweighted windows, top-1 accuracy falls from 42.7% to 39.7%, while outside them it rises from 45.3% to 47.1%, so the two halves of the day exchanged their difficulty along with their labels. The model actively conditions its transition probabilities on the hour tokens, which validates Hypothesis 1.

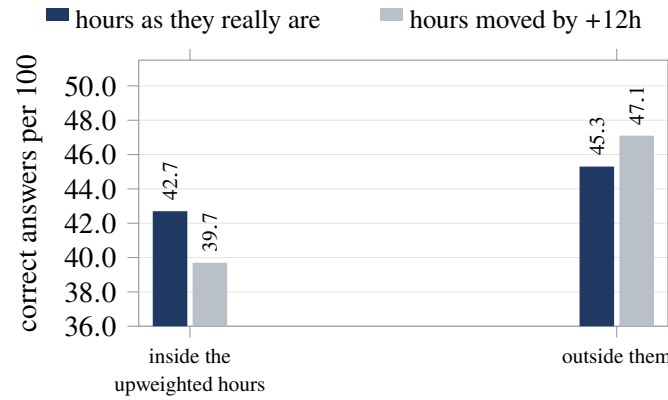


Figure 6.4: The +12h shift swaps the difficulty of the two halves of the day

6.2.2 Evaluation of Weighted Loss and Targeted Sampling

To test Hypothesis 2, three joint modifications were implemented:

1. **Unequal loss weighting:** 18:00–20:00 transitions received a $4\times$ loss multiplier and 04:00–06:00 transitions $2\times$.
2. **Targeted sequence sampling:** training sequences containing a transition-window event were sampled $3\times$ more often, raising their share of the batches from 45.1% to 71.2%.
3. **Full-trajectory evaluation protocol:** evaluation moved from scoring only the final 20% of each trajectory to scoring every event step, which lowers the persistence baseline from 45.5% to 43.1%.

Figure 6.5 and **Table 6.2** evaluate the effect of these adjustments on the 100 held-out subscribers.

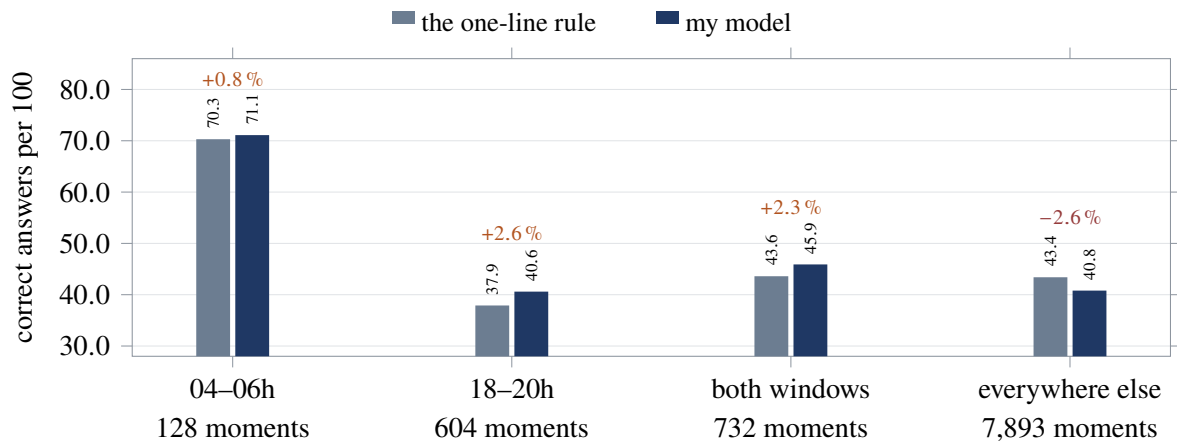


Figure 6.5: Emphasising two stretches of the day: the win, and the price

Table 6.2: The same change, measured everywhere else: seven cuts, seven losses

Past given	Distance before	after	Change	top-3, before → after	top-5, before → after
30 %	−0.9	−1.6	−0.7	61.4 % → 58.8 %	69.1 % → 65.7 %
40 %	−0.6	−1.5	−0.9	61.7 % → 58.8 %	69.0 % → 65.6 %
50 %	−0.7	−1.1	−0.4	61.4 % → 58.8 %	68.6 % → 65.4 %
60 %	−0.9	−1.5	−0.6	61.7 % → 58.7 %	68.7 % → 65.3 %
70 %	−0.3	−1.3	−1.0	61.4 % → 58.5 %	68.6 % → 65.2 %
80 % (<i>ref.</i>)	−0.9	−1.3	−0.4	62.3 % → 59.1 %	70.2 % → 65.4 %
90 %	−2.1	−2.8	−0.7	64.6 % → 60.6 %	72.3 % → 67.7 %

→ Produced by `train_cellid_llm.ipynb`, the targeted-sampling run, evaluated at seven cuts.

6.2.3 Analysis of Empirical Window Misalignment

Each bar pair of **Figure 6.5** is one group of evaluated moments, with its size printed under the label; grey-blue is the persistence baseline and dark blue the model, on exactly the same moments. Within the 18:00–20:00 window (604 moments), the model gains +2.6 % (40.6% vs. 37.9%), and +2.3 % across both windows combined (732 moments; 45.9% vs. 43.6%): the first time in the project that a trained model beat the baseline on a segment chosen in advance. The 04:00–06:00 window rests on only 128 moments and its +0.8 % (71.1% vs. 70.3%) reads as “not worse” rather than as a win. The fourth pair, everything outside the windows, carries 7,893 moments, more than ten times the volume of the gain, and there the model loses −2.6 % (40.8% vs. 43.4%).

Table 6.2 shows that this loss outside the windows is not an artifact of one cut. Each row is a context fraction (“30%” means the first 30% of each day is given as history), and the distances before and after are model minus baseline on the non-target moments. At all seven cuts the model now trails the baseline, by −1.1% to −2.8%, and top-3 and top-5 accuracy fall by 2.6 % to 4.8 %: the correct cell has not slipped from first to second place, it has left the model’s top candidates entirely.

Loss weighting therefore reallocates the model’s optimization focus toward the targeted intervals, a trade worth making only if those intervals are the hard ones. They were not: on full test trajectories, the persistence baseline scores 43.6% *inside* the guessed windows versus 43.4% *outside* them (**Figure 6.5**). Early morning hours (04:00–06:00) are in fact highly regular across all user cohorts (baseline accuracy 49%–64%). As documented by Song et al. [13], nocturnal hours exhibit maximum trajectory regularity, whereas transition difficulty dips around midday and early evening. Globally fixed hours, applied uniformly across all subscribers, fail to account for individual temporal mobility variations. This motivates the behavioral stratification approach presented in Chapter 7.

6.3 Conclusion

This chapter validated the integration of explicit temporal tokens (<H00>–<H23>). Ablation analysis demonstrated that shuffling temporal tokens causes a 3.0 % accuracy drop (**Figures 6.3 and 6.4**), confirming that model predictions depend on temporal sequence ordering.

Conversely, applying globally fixed hourly loss weighting (04:00–06:00 and 18:00–20:00) demonstrated that while loss weighting successfully forces local optimization (+2.6 % at

18:00–20:00 in **Figure 6.5**), applying fixed hours across all subscribers penalizes non-targeted intervals. Because temporal mobility patterns differ significantly across subscribers, effective transition loss weighting requires individual user cohort stratification, as detailed in Chapter 7.

In hindsight, the two experiments of this chapter answer two different questions. The ablation shows that the model reads the hour, since corrupting its order costs more than deleting it. The sampling experiment shows that reading the hour is not enough when the emphasised hours are the wrong ones: the model beat the baseline on a segment chosen in advance (+2.6% at 18:00–20:00), but trailed it at every one of the seven cuts outside the windows (−1.1% to −2.8%), and the guessed windows were no harder than the rest of the day (43.6% against 43.4% for the baseline). The mechanism was right; the hours it was pointed at were not.

7 User Stratification and Per-Group LoRA Adapter Routing

7.1 User Behavioral Clustering Objectives

Chapter 6 established that applying a globally fixed hourly loss weighting across all subscribers is suboptimal because temporal mobility distributions vary significantly between individuals. While certain subscribers commute during early morning hours, others move during late evening periods, and stationary subscribers exhibit minimal transition activity.

To resolve these temporal variations, two research questions were investigated:

1. Can subscribers be classified into distinct behavioral cohorts based on temporal transition profiles, using only historical prefix data available at inference time?
2. Does conditioning model parameter optimization on identified user cohorts improve next-cell prediction accuracy over persistence baselines?

Data Provenance. Analysis evaluates dataset cohort the MOBILE-USER DAY (Table 3.1), fitting behavioral profiles on **400 training subscribers** and evaluating performance on **100 held-out subscribers** across full-trajectory evaluation (persistence baseline score: 43.1%).

7.2 Feature Engineering for Temporal Mobility Profiles

User stratification is established by constructing a four-dimensional temporal mobility vector for each subscriber. Physical base station mast transitions are computed across four daily intervals: night (22:00–07:00), morning (07:00–12:00), afternoon (12:00–18:00) and evening (18:00–22:00).

Two key design constraints govern vector construction:

- **Mast-Level Transition Resolution:** Mobility is evaluated strictly at physical base station mast resolution, excluding intra-site sector handovers (as established in Chapter 3).
- **Inference-Time Causal Integrity:** Temporal mobility profiles are computed strictly from historical prefix segments available up to the prediction moment. Computing profiles on complete daily traces would introduce future information leakage.

7.3 Taxonomy of Behavioral Groups and Per-Group Temporal Hardness

Subscriber trajectories were partitioned into four behavioral clusters using k -means clustering evaluated on historical mobility vectors ($k = 4$). No cluster was labelled in advance: the procedure returns four average shapes, and the group names below are descriptive labels assigned afterwards.

Figure 7.1 details, for each of the four identified behavioral groups ($G0$ to $G3$), the hourly base station transition rate: out of every 100 consecutive records written during a given hour,

how many land on a different mast from the previous one. The percentage attached to each group is its share of subscribers, not of records:

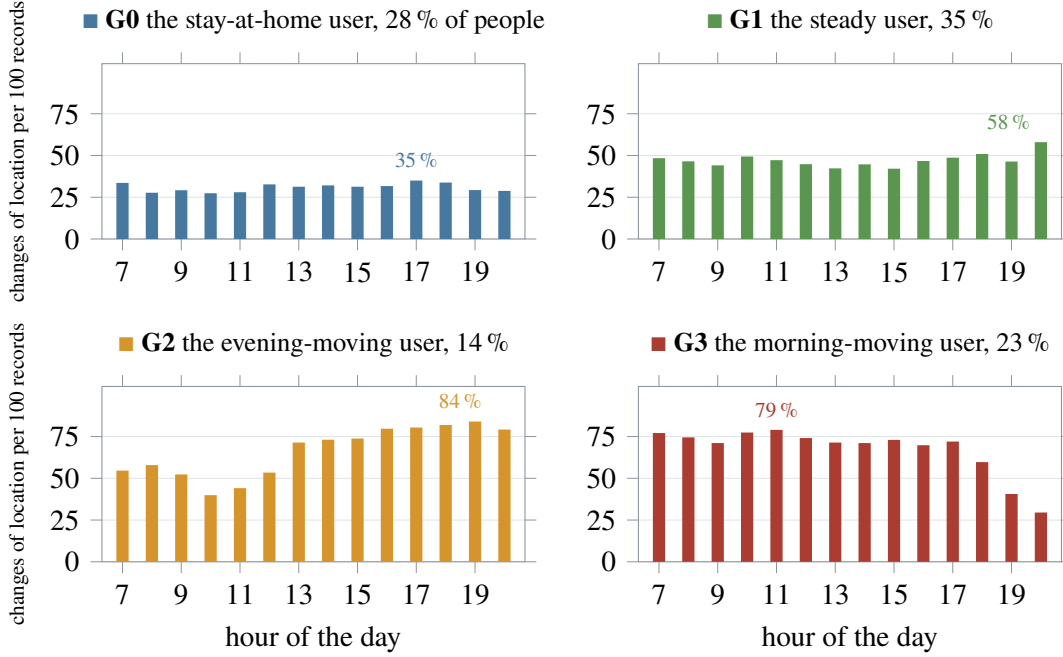


Figure 7.1: The four kinds of user, hour by hour

- G0 (Stationary Subscribers, 28% of population).** Low, uniform transition activity throughout the day (≈ 30 transitions per 100 records, peaking at 35% at 17:00).
- G1 (Regular Mobile Subscribers, 35% of population).** Moderate, uniform transition activity across daylight hours (≈ 45 transitions per 100 records, rising to 58% at 20:00).
- G2 (Evening-Mobile Subscribers, 14% of population).** Low morning activity transitioning to peak mobility during late afternoon and evening hours (≈ 84 transitions per 100 records at 19:00).
- G3 (Morning-Mobile Subscribers, 23% of population).** Peak mobility during early morning hours (≈ 79 transitions per 100 records before 12:00), declining to 29.5% during evening hours.

The inverse temporal trends of *G2* and *G3* explain why globally fixed temporal loss weighting was ineffective: the two groups are near mirror images comprising 37% of the total population, so any globally fixed temporal window that is hard for *G2* is easy for *G3*.

Figure 7.2 validates the efficacy of behavioral stratification. **Figure 7.2.a** demonstrates an exact inverse relationship between physical group mobility (orange bars) and persistence baseline predictability (grey-blue bars): stationary *G0* subscribers change location 33% of the time and yield a 56.3% baseline accuracy, whereas morning-mobile *G3* subscribers change location 52% of the time and yield a baseline accuracy of only 17.8%. The two bars use different scales, so each colour is read against its own neighbours; the two rankings are nonetheless exact opposites. Overall, cluster assignments account for 72% of predictability variance across held-out test subscribers.

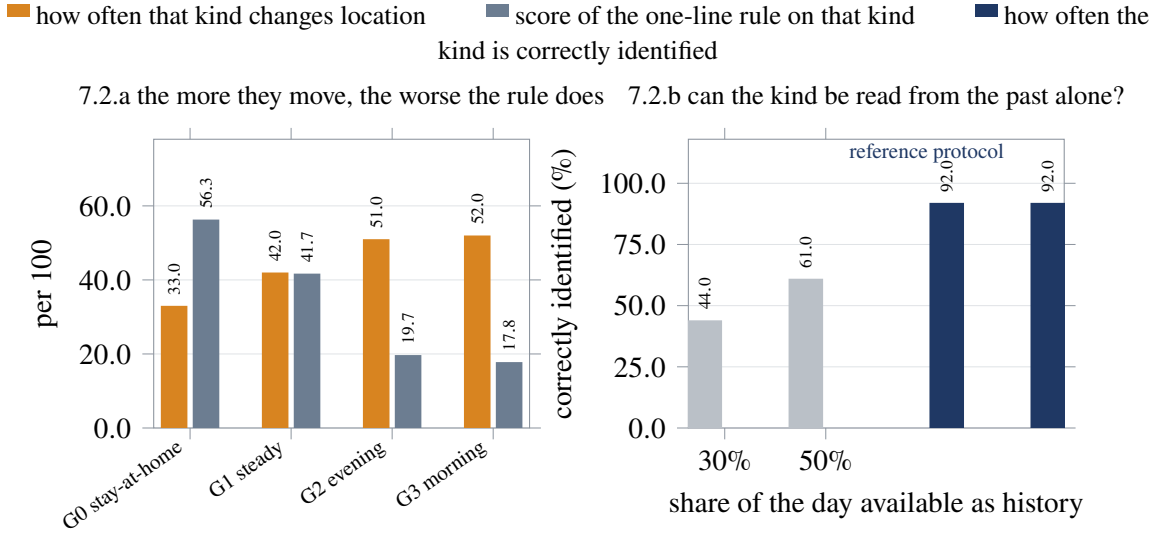


Figure 7.2: The kind of user predicts difficulty, and can be read from the past

↔ Produced by `train_cellid_llm.ipynb`, the group-detection and per-group baseline cells.

Figure 7.2.b evaluates inference-time causal integrity by measuring how accurately user group assignments can be derived strictly from historical prefix segments. Given an 80% prefix context, historical assignment matches full-day assignment for 92 out of 100 test subscribers (92% accuracy). Below a 60% context fraction, detection accuracy degrades rapidly (61% at 50% context, 44% at 30% context): the assignment is unreliable precisely for subscribers who have only just appeared, and early-stage trajectory routing would require alternative short-prefix classification methods.

7.3.1 Data-Driven Derivation of Critical Mobility Windows

Utilizing group-specific temporal profiles from **Figure 7.1**, critical transition windows were derived per group by identifying hourly intervals where the group’s persistence baseline accuracy falls furthest below its daily average.

Table 7.1 lists the derived temporal transition windows and the loss cost multiplier applied to a mistake inside each of them. For stationary *G0* subscribers, no transition windows are upweighted since mobility remains uniformly low; for *G2* and *G3*, the derived windows are mutually disjoint, reflecting their opposing diurnal routines. These windows successfully isolate high-difficulty moments: baseline accuracy inside them drops to **19.0%** compared to **46.4%** outside, a **27.4 % separation**. The globally fixed heuristic windows (04:00–06:00 and 18:00–20:00) separated baseline difficulty by only 0.2 % (43.6% vs. 43.4%), and were misplaced rather than merely imprecise: 04:00–06:00 is easier than average for all four groups. Data-driven group stratification is therefore necessary to identify true transition windows.

Table 7.1: Hours derived per kind of user, and the hard moments they land on

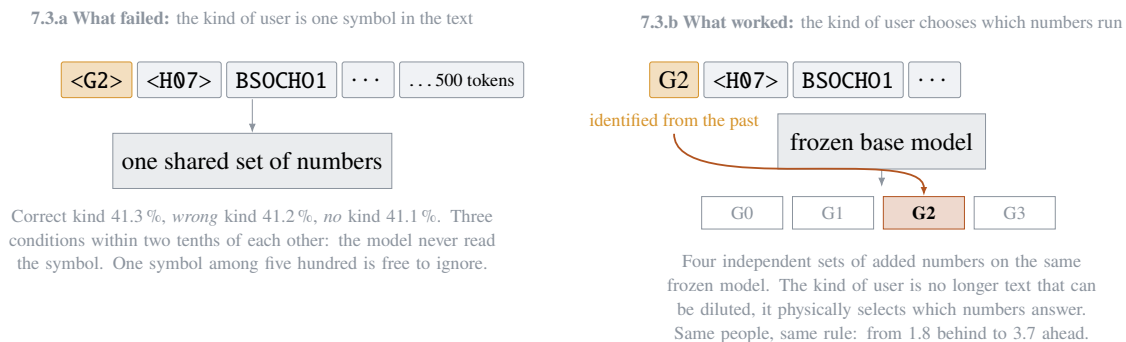
Kind of user	Hours derived from the data (cost multiplier)	What it says
G0 stay-at-home user	<i>none</i>	a flat day has no hours worth emphasising
G1 steady user	18–19h $\times 1.4$, 20–21h $\times 1.7$	mild, and late in the day
G2 evening-moving user	13–14h $\times 1.7$, 16–21h $\times 2.5$	the five-hour block was later narrowed to 18–21h
G3 morning-moving user	07–08h $\times 2.0$, 10–13h $\times 2.3$	the mirror image of G2, and almost disjoint from it
Do those hours actually mark the hard moments? Score of the one-line rule, inside versus outside		
The guess of Ch. 6, 04–06h / 18–20h		43.6 inside vs 43.4 outside: a difference of 0.2 , so it marks nothing
The hours derived per kind of user		19.0 inside vs 46.4 outside: a difference of 27.4 , the genuinely hard hours

7.4 Model Conditioning and Per-Group Adapter Architecture

7.4.1 Diagnostic Analysis of Textual Group Token Conditioning

Initial conditioning experiments inserted explicit text tokens (<G0> through <G3>) at the beginning of input sequence prefixes. However, this textual conditioning model achieved 41.3% accuracy against a 43.1% persistence baseline (-1.8 %).

Diagnostic ablation revealed the root cause: evaluating the model with correct group tokens (41.3%), incorrect group tokens (41.2%), or omitted group tokens (41.1%) yielded virtually identical accuracies across a narrow 0.2 % range. **Figure 7.3** compares this input prefix token conditioning (**Figure 7.3.a**) against physical parameter routing via separate LoRA adapters (**Figure 7.3.b**), with the same subscribers, protocol and weighted hours in both. As **Figure 7.3.a** illustrates, self-attention layers diluted the single prefix group token across extended 500-token trajectory sequences, allowing the model to ignore the conditioning signal.

**Figure 7.3:** The kind of user as a piece of text, and as a switch

7.4.2 Implementation of Per-Group LoRA Adapter Routing

To enforce group conditioning, per-group LoRA adapter routing was implemented (**Figure 7.3.b**):

1. Base language model parameters remain frozen.
2. Four independent LoRA adapter modules are instantiated, corresponding to $G0$ – $G3$.
3. Input trajectories are dynamically routed to the LoRA adapter corresponding to the subscriber’s historical cluster assignment.

Physical parameter routing eliminates text token dilution by directing input trajectories to specialized subgroup weights, enabling collaborative parameter sharing among subscribers with identical mobility profiles. Four adapters also carry four times the trainable capacity of one, and the control that would separate capacity from stratification was not run.

7.4.3 Performance Evaluation of Adapter Routing

Evaluating per-group LoRA adapter routing on active mobility test cohort the `MOBILE-USER DAY` yields an overall top-1 accuracy of **46.8%** against a persistence baseline of **43.1%**, a statistically significant **+3.7 % gain**. **Figure 7.4** breaks it down with one bar pair per user cohort ($G0$ – $G3$), baseline and model scored on exactly the same moments, and their plain average on the far right. Accuracy improvements rank in the exact reverse order of baseline group predictability: the largest gains manifest in high-mobility groups ($G3$ with +6.4 %, 24.2% vs. 17.8%; $G2$ with +5.4 %, 25.1% vs. 19.7%; $G1$ with +2.3 %, 44.3% vs. 42.0%), whereas the low-mobility group $G0$ performs close to its baseline (+0.8 %, 56.8% vs. 56.0%). The +3.7 % counts each group once; weighted by group size it is +3.3 %, the figure an operator would experience, because 28% of subscribers sit in $G0$.

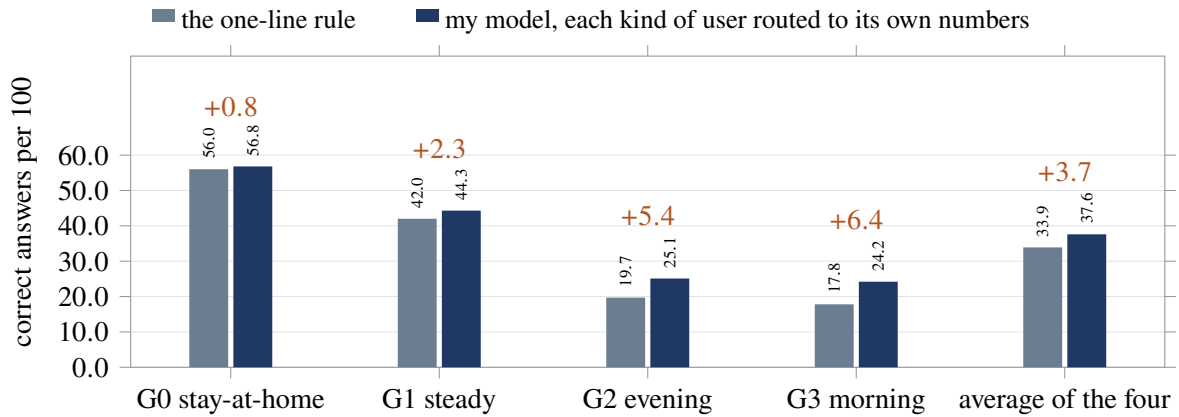


Figure 7.4: The gain, kind of user by kind of user

↪ Produced by `train_cellid_llm.ipynb`, the per-group adapter-routing run.

7.5 Conclusion

Integrating user stratification with per-group LoRA adapter routing produced the first experimental model to surpass the persistence baseline across complete held-out evaluation trajectories (+3.7 % in **Figure 7.4**). Diagnostic analysis confirmed that routing trajectories to specialized subgroup parameters resolves text token dilution (**Figure 7.3.b**) while enabling collaborative parameter sharing among subscriber cohorts.

This result rests on two conditions established in this chapter. The kind of user must be readable from the past alone, which holds for 92 of 100 test subscribers with an 80% prefix but degrades quickly below a 60% prefix; and it must reach the model through its parameters rather than through a prefix token, which the model ignored (41.3%, 41.2% and 41.1% with the correct, a wrong or no token). The gain is also uneven: +6.4 % for $G3$ and +5.4 % for $G2$, but only +0.8 % for $G0$. Finally, four adapters carry four times the trainable capacity of one, and the control that would separate capacity from stratification was not run, so part of the gain may come from capacity alone.

8 Scaling Analysis: Impact of Cohort Size on Predictability

8.1 Scaling Analysis Objectives

Stratifying 400 training subscribers across four behavioral user groups leaves smaller clusters (such as *G2*, comprising 14% of the cohort) with approximately 56 training trajectories. Estimating robust subgroup transition probabilities from limited samples presents empirical bounds.

This observation leads to the central research question evaluated in this chapter: how does increasing subscriber cohort size impact next-cell prediction accuracy, and at what scale does performance saturate?

Data Provenance. Evaluation spans three dataset cohorts from **Table 3.1**: 400 subscribers (the *MOBILE-USER DAY*, 100 held-out subscribers, baseline 43.1%), 900 to 1,900 subscribers (the *2,000-USER DAY*, 100 held-out subscribers, baseline 48.2%), and 5,000 to 21,000 subscribers (the *ELEVEN-DAY SET*, **1,000 held-out subscribers**, baseline 50.2%). Because baseline scores vary across evaluation cohorts, performance is evaluated strictly using accuracy deltas relative to each cohort’s one-line baseline (“repeat the last antenna”) rather than absolute accuracy percentages, ensuring robust comparisons across varying test subscriber pools.

8.2 Experimental Setup for Cohort Size Sweep

The cohort scaling evaluation sweeps training population size across eight nested steps (400, 900, 1,400, 1,900, 5,000, 10,000, 15,000, and 21,000 subscribers) while holding model architecture, LoRA adapter configuration, hour tokens, and per-group routing constant. Nested sampling ensures that larger cohorts strictly expand smaller cohorts without introducing population composition shifts. Across training sizes, training log counts increase from 369,451 to 1,538,379 events.

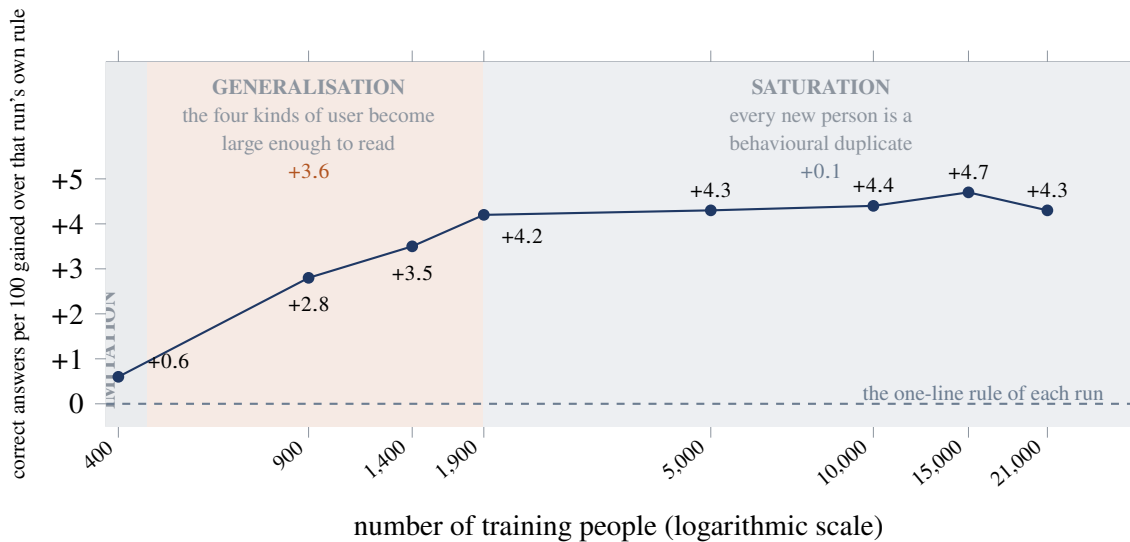


Figure 8.1: Eight training sizes: the gain climbs, then stops

↪ Produced by `train_cellid_final.ipynb`, the eight-tier scaling notebook.

Figure 8.1 plots prediction accuracy delta relative to baseline against training cohort size on a logarithmic scale, so that equal distances are equal multiples. The vertical axis is not a score but the distance to the baseline on each run's own test subscribers, and the dashed line at zero marks a model indistinguishable from repeating the last antenna. The empirical curve exhibits three distinct regimes:

1. **Imitation regime (400 subscribers):** At 400 training subscribers, the model achieves a marginal +0.6 % delta over baseline, failing to capture group-level transition dynamics.
2. **Generalization regime (400 to 1,900 subscribers):** Accuracy delta climbs rapidly through +2.8 % (900 subscribers) and +3.5 % (1,400 subscribers) to +4.2 % (1,900 subscribers), capturing essentially the entire benefit derived from dataset volume expansion.
3. **Saturation regime (1,900 to 21,000 subscribers):** A ten-fold expansion in subscriber scale from 1,900 to 21,000 subscribers yields negligible additional gains (+4.3 % at 5,000, +4.4 % at 10,000, +4.7 % at 15,000, and +4.3 % at 21,000 subscribers), and the best size (15,000) is followed by the worst (21,000): performance growth halts rather than merely slowing down.

Two important experimental caveats govern this scaling trend. First, because test subscriber pools vary across cohort sizes, the small step across the boundary between two test pools is not a measurement. Second, all scaling steps were evaluated under a constant compute budget (Section 8.3.1), so the defensible claim is that 1,900 subscribers suffice *at this budget*.

8.2.1 Statistical Validation of Performance Saturation

To establish whether performance plateauing above 2,000 subscribers reflects true saturation rather than random variance, paired bootstrap resampling [3] was conducted across 2,000 resamples of the 1,000 held-out subscribers. In each bootstrap iteration, 1,000 subscribers were sampled with replacement and both candidate models were scored on the identical subscriber draw, canceling evaluation sample variance.

Table 8.1 presents the resulting 95% confidence intervals across cohort scale transitions. For all transitions above 5,000 subscribers, the 95% confidence interval spans zero (e.g., net delta between 5,000 and 21,000 subscribers is +0.04 %, 95% CI $[-0.43, +0.51]$): the claim is not that the gain is small, but that no gain was measured. Fitting a log-linear scaling law across all eight data points yields a minimal slope of +0.18 % per decade of subscribers, accounting for only 12% of variance ($R^2 = 0.12$).

Table 8.1: No step up the ladder is real: every interval contains zero

Step in training people	Measured difference	Range it could really be	Real gain?
5,000 → 10,000	+0.20 %	-0.21 ... + 0.64	no
10,000 → 15,000	+0.09 %	-0.32 ... + 0.52	no
15,000 → 21,000	-0.25 %	-0.72 ... + 0.20	no
5,000 → 21,000 (<i>total</i>)	+0.04 %	-0.43 ... + 0.51	no

↪ Produced by `train_cellid_final.ipynb`, the paired-bootstrap cells, 2,000 resamples.

Table 8.2 confirms that performance saturation holds uniformly across six complementary evaluation metrics. Across four training sizes (5,000 to 21,000 subscribers), top-1 accuracy (54.5% to 54.9%, spread 0.4 %), top-3 accuracy (71.4% to 72.2%, spread 0.2 %), top-5 accuracy (78.6% to 79.3%, spread 0.4 %), cold-start accuracy (51.4% to 52.6%, spread 0.5 %), median subscriber accuracy (49.1% to 50.3%, spread 1.2 %), and worst-decile accuracy (13.1% to 14.4%, spread 1.3 %) show no systematic improvement; the 1,900-subscriber reference column, marked [†], is measured on the 2,000-USER DAY against a 48.2% baseline and is excluded from the spread. Crucially, under cold-start conditions (evaluating only the first 30% of a subscriber’s daily history), the model achieves its highest edge over baseline (+4.6 % vs. +4.3 % at 80% context history), demonstrating that model capabilities are most effective when user history is sparse, and this advantage remains identical at 5,000 and 21,000 subscribers.

Table 8.2: Six ways of measuring, and all six say the same thing

Way of measuring	1,900 [†]	5,000	10,000	15,000	21,000	Spread
Top-1, the headline	52.4 %	54.5 %	54.6 %	54.9 %	54.5 %	0.4 %
Top-3	71.4 %	72.0 %	72.1 %	72.1 %	72.2 %	0.2 %
Top-5	78.6 %	78.9 %	78.9 %	78.9 %	79.3 %	0.4 %
Top-1 on a cold start, 30 % of the day	51.4 %	52.3 %	52.6 %	52.3 %	52.1 %	0.5 %
The middle person		49.1 %	49.2 %	49.2 %	50.3 %	1.2 %
The worst tenth of people		14.4 %	13.1 %	14.4 %	13.1 %	1.3 %

↪ Produced by `train_cellid_final.ipynb`, the six-metric evaluation of the four tiers.

Furthermore, **Table 8.3** evaluates performance across context history fractions (30% to 90% of daily observations), comparing 900 and 1,900 subscriber models graded on the same 100 held-out subscribers. The 1,900-subscriber model leads the baseline by +4.3 % at 30% context and +4.2 % at 80%, and across all context fractions expanding training cohort size from 900 to 1,900 subscribers yields a consistent gain between +0.9 % and +1.7 %, establishing that dataset scale improvements operate independently of context history length.

Table 8.3: The gain does not come from giving the model a long history

Past given	Moments	The rule	900 people	1,900 people	Difference
30 %	5,491	47.1 %	50.0 % (+2.9)	51.4 % (+4.3)	+1.4
40 %	4,703	46.8 %	49.9 % (+3.1)	51.4 % (+4.6)	+1.5
50 %	3,922	46.6 %	50.1 % (+3.5)	51.5 % (+4.9)	+1.4
60 %	3,142	47.0 %	50.3 % (+3.3)	51.2 % (+4.2)	+0.9
70 %	2,359	47.3 %	50.5 % (+3.2)	51.6 % (+4.3)	+1.1
80 % (<i>ref.</i>)	1,570	48.2 %	51.0 % (+2.8)	52.4 % (+4.2)	+1.4
90 %	781	50.3 %	51.9 % (+1.6)	53.6 % (+3.3)	+1.7

↪ Produced by `train_cellid_final.ipynb`, the context-fraction evaluation at the smaller tiers.

8.2.2 Sensitivity Analysis: User Typology vs. Data Scale

Evaluating the 21,000-subscriber model across behavioral cohorts reveals substantial variance: top-1 accuracy reaches 62.8% for stationary subscribers (*G0*, 22% of cohort), 53.6% for regular mobile subscribers (*G1*, 37%), 43.0% for evening-mobile subscribers (*G2*, 22%), and 40.4% for morning-mobile subscribers (*G3*, 18%). This establishes a performance spread of **22.4 % across user types**, compared to a marginal **0.4 % spread across dataset scale** (from 1,900 to 21,000 subscribers in **Figure 8.1**).

Subscriber mobility typology exerts a 50× stronger impact on prediction accuracy than scaling training cohort size beyond 2,000 subscribers.

8.3 Mechanistic Analysis of Saturation Thresholds

Figure 8.2 details two physical mechanisms underlying performance scaling and saturation:

1. **Subgroup Sample Threshold (Figure 8.2.a):** Tracking subscriber counts in the smallest behavioral cohort (*G2* evening-mobile subscribers, 14% population share) reveals 56 members at 400 subscribers (+0.6 % gain), 126 members at 900 subscribers (+2.8 % gain), 196 members at 1,400 subscribers (+3.5 % gain), and 266 members at 1,900 subscribers (+4.2 % gain). Crossing approximately 100 members in the smallest subgroup provides sufficient sample volume to robustly estimate subgroup-specific transition matrices, explaining the performance jump between 400 and 900 subscribers.
2. **Spatial Vocabulary Coverage Saturation (Figure 8.2.b):** Each square is one of the 100 antennas the test subscribers actually use; the grid shows that 99 cells are already observed at 400 subscribers, and 100% coverage is reached by 5,000 subscribers. Because cell vocabulary coverage saturates and the underlying behavioral transition patterns (**Figure 7.1**) remain invariant, expanding cohorts beyond 5,000 subscribers introduces redundant trajectories across previously seen cell graphs rather than novel mobility structures.

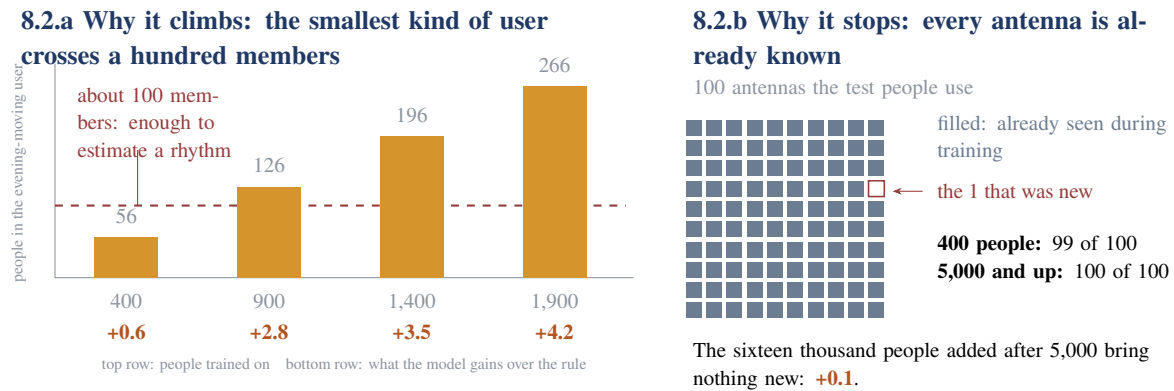


Figure 8.2: The reason the curve climbs, and the reason it stops

↪ Produced by `train_cellid_final.ipynb`, the eight-tier scaling notebook; the member counts are the group shares of Chapter 7 applied to each size.

These two mechanisms fit the curve without proving it: the alternative reading, that the compute budget rather than the population ran out, is set out in Section 8.3.1.

8.3.1 Methodological Constraints and Scope Boundaries

Scaling experiments maintained constant compute step budgets (8,000 gradient update steps per run). At 5,000 subscribers (5,379 training sequences), epoch coverage reaches 1.5 passes. At 21,000 subscribers (22,522 sequences), 8,000 steps correspond to 36% single-pass epoch coverage. Consequently, reported saturation bounds hold under fixed compute step constraints and fixed parameter adapter capacity.

8.4 Evaluation of Trajectory Length Equalization

An alternative preprocessing approach was evaluated: equalizing subscriber trajectory lengths by oversampling short trajectories to a uniform 200 records per subscriber, addressing data imbalances where the top 25% longest trajectories comprise 44% of training events.

Figure 8.3 evaluates trajectory equalization, each run shown against its own baseline: trajectory stretching yields an apparent top-1 accuracy of 74.9%, representing the highest absolute score recorded. However, evaluating the baseline rule on the same equalized trajectories yields 76.5%, placing the fine-tuned model –1.6 % behind baseline.

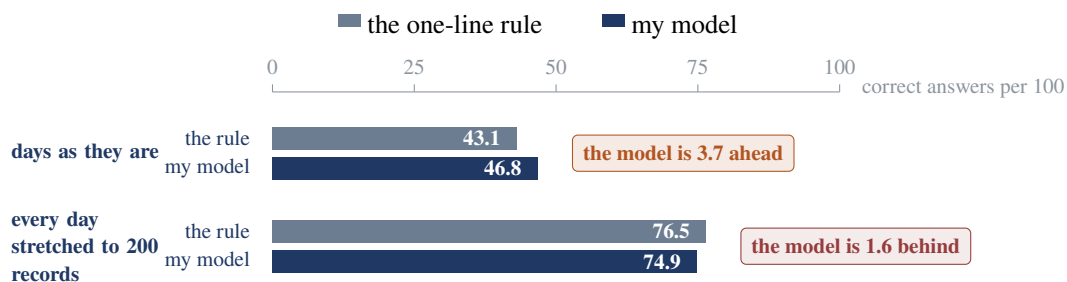


Figure 8.3: Stretching every day to 200 records: a high score, a worse result

Figure 8.4 exposes the underlying breakdown: oversampling short trajectories inserts 2,262 pure copy moments (out of 4,000 total test points) where target cell IDs directly

replicate preceding tokens. On these copy moments, both baseline and model achieve 100% accuracy, artificially inflating baseline scores by +33.4 %. On the remaining 1,738 true decision moments where movement occurs, baseline accuracy is 45.9% while model accuracy drops to 42.2% (a net −3.7 % degradation; the model bar is derived from the reported 74.9% assuming it never misses a copy, and stays below the baseline at either end of that assumption). Moreover, zero-movement copy step repetition compresses inter-subscriber accuracy variance by a factor of 2.3, destroying the behavioral variance critical for subgroup stratification. Equalizing trajectory lengths distorts empirical transition distributions and degrades true decision quality.

2,262 pure copies	1,738 real decisions
The antenna asked for is a copy of the one just before, so anything that repeats is right: 100 out of 100 , for the rule and for the model alike. This block is the whole of the rise in Figure 8.3.	The moments where somebody had to be predicted. Here the rule is right 45.9 times in 100 and my model 42.2 : 3.7 behind , on exactly the moments that matter.

Figure 8.4: The 4,000 graded moments once the days have been stretched

8.5 Conclusion

Cohort scaling analysis demonstrates that next-cell prediction accuracy saturates at approximately 2,000 subscribers under parameter-efficient fine-tuning (+4.2 % over baseline in **Figure 8.1**). Expanding training cohorts from 2,000 to 21,000 subscribers yields negligible accuracy gains (+0.04 %, $p > 0.05$ in **Table 8.1**). Furthermore, individual subscriber mobility typology accounts for a 22.4 % accuracy spread, demonstrating that individual user characterization exerts a far stronger influence on predictability than raw data scale.

Two mechanisms account for the shape of the curve: the gain rises while the smallest kind of user, *G2*, grows past about 100 members (56 at 400 subscribers, 126 at 900), and it stops once the antennas used by the test subscribers are all known, 99 of 100 of them already at 400 subscribers. The saturation was measured under a fixed budget of 8,000 steps, so at 21,000 subscribers the model sees only 36% of its data once; it is a statement about this budget, not about every budget. Finally, stretching every day to 200 records produced the highest absolute score of the project (74.9%) and a worse result, 1.6 % behind the baseline, a reminder that only the distance to the baseline counts.

9 Conclusion

The central objective of this research was to determine whether a parameter-efficient fine-tuned language model trained on anonymized mobile network logs can accurately forecast a subscriber’s next cell attachment. Over nine weeks of experimental investigation across eleven iterative configurations, the empirical results confirm that fine-tuned language models achieve superior prediction accuracy compared to strong persistence baselines, with optimal performance governed by specific structural user conditioning mechanisms.

9.1 Summary of Key Findings

Three primary empirical findings summarize the contributions of this research, addressing the central research questions, as synthesized in **Figure 9.1**. Its three cards are ordered by how firmly each result is established: the +4.3 of Card 1 is one measurement repeated once on a second cohort, the 1,900 of Card 2 rests on eight training sizes and a resampling test, and the 22.4 of Card 3 recurs in every per-user breakdown.

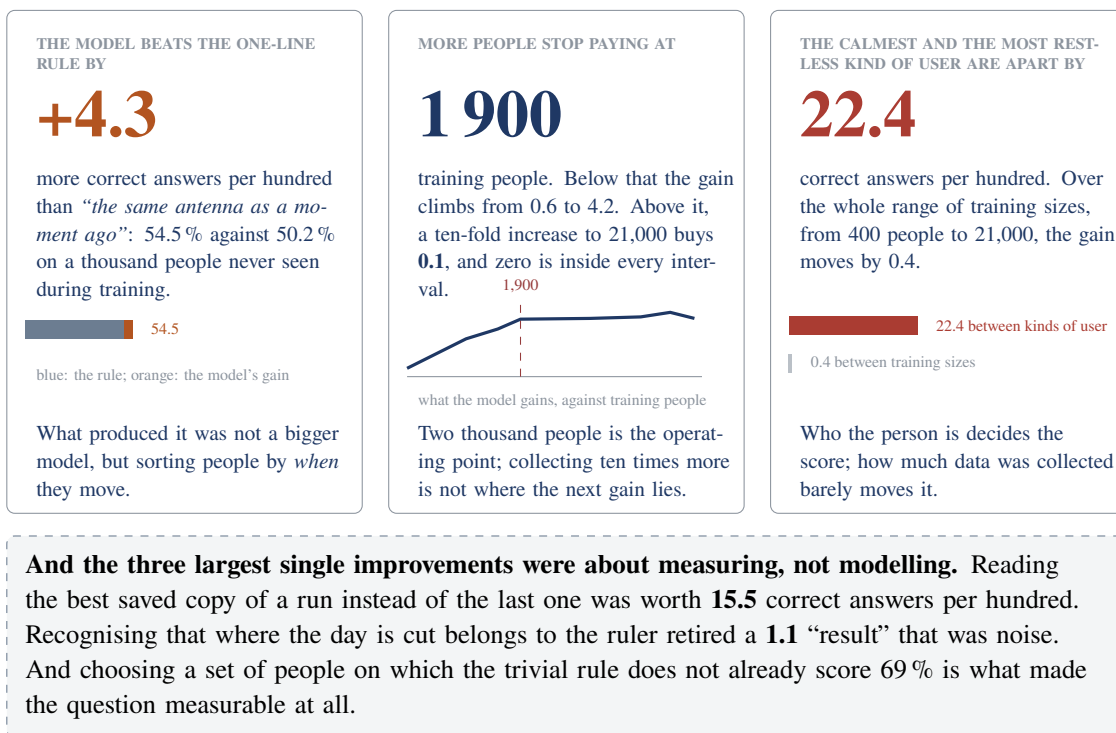


Figure 9.1: The whole report in three numbers

↔ Produced by `train_cellid_final.ipynb`, with every number traced to its run in Table B.2.

Demonstrated Baseline Outperformance via Behavioral Conditioning. A fine-tuned Qwen2.5-0.5B language model achieves a **+4.3 % accuracy gain** over the persistence baseline on active mobility evaluation cohorts (54.5% vs. 50.2% top-1 accuracy on 1,000 held-out subscribers, Card 1 of **Figure 9.1**). Surpassing the persistence baseline required user behavioral stratification (*G0–G3*) and per-group LoRA adapter routing, which increased model accuracy from -1.8 % behind the baseline to +3.7 % ahead on baseline active mobility splits. Rather than increasing general parameter capacity, temporal behavior profiling represents the

primary driver of performance.

Data Cohort Scaling Saturation at 2,000 Subscribers. Scaling training cohort size yields significant accuracy improvements up to 2,000 subscribers (+4.2 % over baseline, Card 2 of **Figure 9.1**). Expanding training cohorts further up to 21,000 subscribers yields no statistically significant improvement (+0.04 %, $p > 0.05$ with zero contained in every bootstrap confidence interval), establishing 2,000 subscribers as the empirical operating threshold for training set construction. Collecting ten times more data is not where future modeling gains lie.

Dominance of Subscriber Typology over Data Volume. Individual subscriber mobility characteristics represent the primary determinant of trajectory predictability. Prediction accuracy across behavioral user groups spans a **22.4 % range** (62.8% for stationary *G0* vs. 40.4% for mobile *G3*, Card 3 of **Figure 9.1**), outweighing the impact of dataset scaling beyond 2,000 subscribers (0.4 %) by a factor of over fifty.

Impact of Methodological and Measurement Rigor. Across experimental iterations, three methodological measurement corrections exerted a larger impact on reported results than individual architectural modifications:

1. **Validation Checkpoint Selection:** Restoring model weights from the best validation loss checkpoint rather than the final step recovered **+15.5 %** in top-1 accuracy.
2. **Evaluation Boundary Alignment:** Aligning daily sequence truncation rules between model and baseline eliminated a **1.1 %** evaluation artifact caused by sequence boundary noise.
3. **Cohort Benchmark Framing:** Transitioning evaluation from highly stationary subscribers (where persistence baselines reach 69%) to active mobile subscriber cohorts enabled meaningful empirical discrimination of model predictive capabilities.

9.2 Experimental Limitations and Unresolved Tasks

Three unresolved technical validation steps are identified for future verification:

1. **Cross-Cohort Model Rescoring:** Rescoring earlier experimental checkpoints across the unified 1,000 held-out evaluation cohort to place all iteration metrics on a single standardized baseline axis.
2. **Multi-Seed Parameter Variance Estimation:** Evaluating multi-seed training runs to quantify variance across stochastic optimization initializations.
3. **LoRA Adapter Capacity Control:** Training a single LoRA adapter with equivalent overall parameter capacity to isolate the contribution of adapter capacity expansion versus group routing.

In addition, hardware constraints limited scaling sweeps to a fixed gradient step budget (8,000 steps), leaving full multi-epoch training on 21,000 subscribers for future evaluation.

9.3 Perspectives and Future Work

Short-Prefix User Stratification. While per-group adapter routing yields significant accuracy gains, identifying user group assignment ($G0$ – $G3$) currently relies on observing extended historical trajectory prefixes. Developing short-prefix classification algorithms would allow early-stage trajectory routing for incoming subscribers.

Multi-Day Trajectory Personalization. Expanding trajectory sequence windows beyond single 24-hour periods to multi-day trajectories would enable capturing weekly subscriber routine patterns and individual personalization.

Cross-City Generalization. Evaluating model transferability across denser urban wireless networks with higher base station densities represents a key direction to test spatial generalization.

9.4 Executive Summary

Parameter-efficient fine-tuning of language models on single-day mobile network logs achieves superior next-cell forecasting accuracy over persistence baselines (+4.3 % top-1 accuracy gain in **Figure 9.1**). Training dataset scaling saturates at approximately 2,000 subscribers, beyond which individual subscriber mobility typology exerts the dominant influence on predictability (22.4 % spread). Stratifying subscribers by temporal mobility profiles and routing input sequences to specialized LoRA adapters represents the primary architectural driver of trajectory forecasting performance, while rigorous evaluation design and checkpoint selection provide the empirical foundation for robust trajectory modeling.

A Timeline of the eleven iterations

The body is organised by question rather than by date. This appendix restores the chronology. **Figure A.1** gives one entry per iteration, with the question it asked, what changed, and the number it produced.

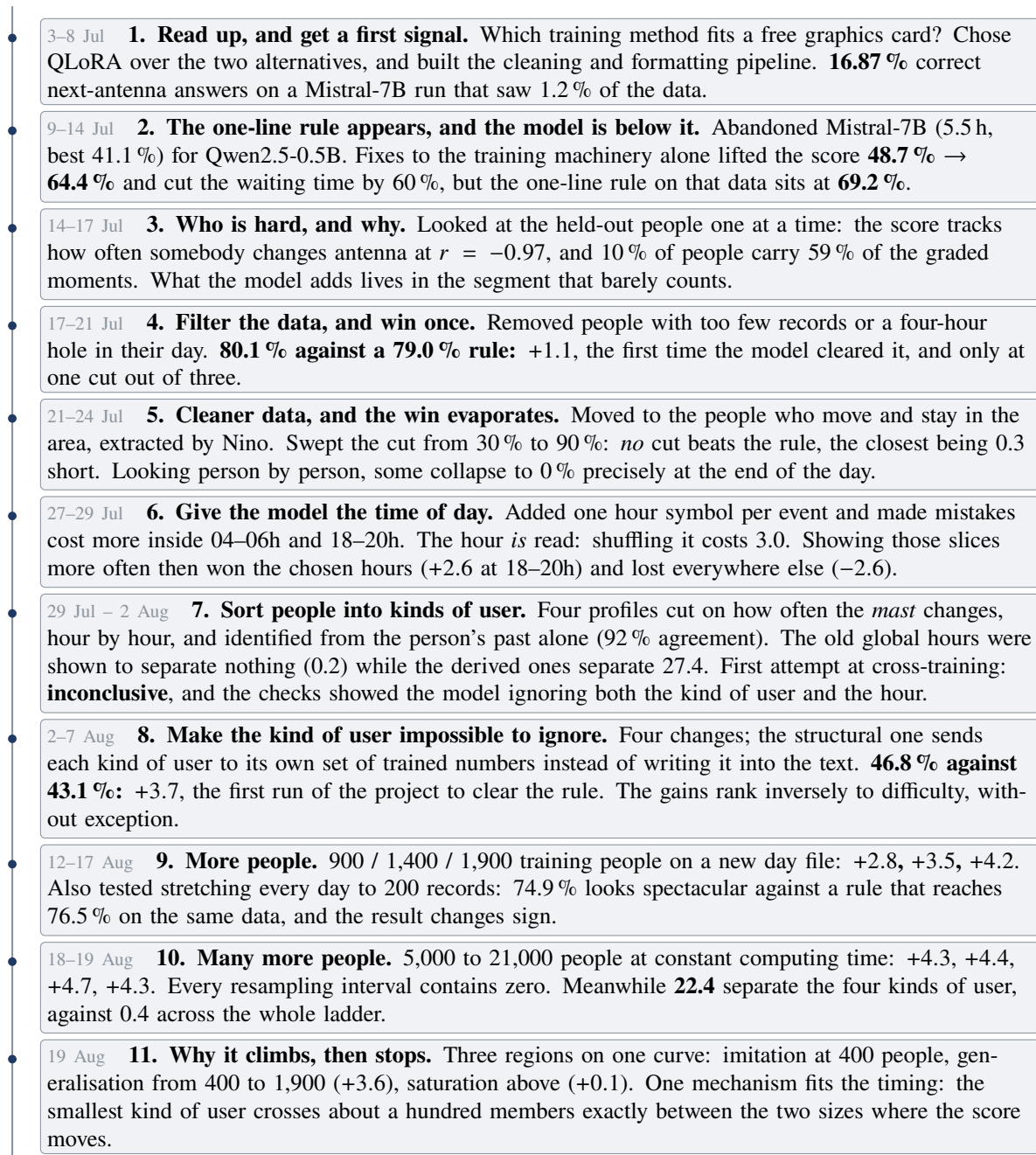


Figure A.1: Timeline of the eleven iterations

↪ Produced by `train_cellid_llm.ipynb`, iterations 1–8; iterations 9–11 come from `train_cellid_final.ipynb` and `train_cellid_week.ipynb`, and every number is traced in Table B.2.

Performance across the iterations of **Figure A.1** must be evaluated strictly using accuracy deltas relative to the one-line persistence rule (“repeat last antenna”) rather than absolute accuracy percentages. Because subscriber filtering criteria and evaluation cohorts changed four times across the nine-week investigation, absolute scores fluctuate significantly: Iteration 2 reached 64.4% top-1 accuracy but lagged –4.8 % behind its 69.2% baseline, whereas

Iteration 8 reached only 46.8% while clearing its 43.1% baseline by +3.7 %. Across the eleven iterations, five early configurations operated below baseline, Iteration 4 produced an unverified transient gain (+1.1 %), and sustained baseline outperformance (+3.7 % to +4.3 %) was established from Iteration 8 onward following the introduction of behavioral user stratification and per-group LoRA adapter routing.

B Reproducibility

To ensure full experimental reproducibility, this appendix documents the hyperparameter configurations, data provenance mappings, and core decision-encoding parameters across all eleven experimental iterations.

Table B.1 records the exact hyperparameter settings across four chronological run families (Iter. 1, Iter. 2–6, Iter. 7–9, and Iter. 10–11). All unlisted hyperparameters were held strictly constant across runs, ensuring that performance variations stem exclusively from data preprocessing, sequence formatting, or adapter conditioning mechanisms rather than hyperparameter tuning artifacts. Key parameters include max pass caps (20 epochs) with early stopping (patience 4), implementing the validation checkpoint selection protocol established in **Chapter 4**, and transition sampling emphasis shifting from global hours to subgroup-specific temporal profiles (**Chapters 6** and **7**).

Table B.1: The training settings of each family of runs

	Iter. 1	Iter. 2–6	Iter. 7–9	Iter. 10–11
Base model	Mistral-7B-v0.1	Qwen2.5-0.5B-Instruct	Qwen2.5-0.5B-Instruct	Qwen2.5-0.5B-Instruct
Loading	4-bit (QLoRA)	bf16 / fp32	bf16 / fp32	bf16 / fp32
LoRA rank r	16	16	16	16
LoRA α	32	32	32	32
LoRA dropout		0.1	0.1	0.1
Learning rate	default	1.5×10^{-4}	1.5×10^{-4}	1.5×10^{-4}
Weight decay		0.01	0.01	0.01
Max passes	100 steps	20	20	20 (cosine horizon)
Early stopping		patience 8, then 4	patience 4	4 evaluations
Eval. sequence length		512 tokens	512 tokens	512 tokens
Training window		128 events, stride 64	128 events, stride 64	128 events, stride 64
Context fraction		0.8	0.8	0.8
Added symbols	antenna names	369 antennas	369 antennas + 24 hours	369 antennas + 24 hours
Transition sampler		$\times 3$ from iter. 6	$\times 3$	$\times 3$
Emphasised hours		global (4–6, 18–20)	per kind of user	per kind of user
Kinds of user			4, read from the past	4, refitted at each size
Stopping protocol	fixed steps	passes	passes	1,000 optimiser steps
Seed		42	42	42
Hardware	Colab T4	Colab T4	Colab T4	Colab T4, 42 min/size

Table B.2 maps every headline result quoted throughout the report to its precise iteration number, underlying dataset split (**Table 3.1**), and notebook execution run. Because dataset filtering and test subscriber cohorts evolved across project stages, comparing absolute scores across rows using different datasets is invalid; meaningful comparisons rely strictly on baseline deltas computed relative to each dataset’s dedicated persistence benchmark.

Table B.2: The origin of each number quoted in the report

Number quoted in the report	Data	Iter.	Which run produced it
16.87 % first signal	RAW DAY	1	single-notebook Mistral-7B QLoRA run, 100 steps
48.7 % $\rightarrow$ 64.4 %	RAW DAY	2	same notebook, keeping the best saved copy
$r = -0.971$ on the switch rate	RAW DAY, 500 people	2	per-person analysis of the 1,173-person grading
+1.1 at the 80/20 cut	FILTERED DAY	4	training run on the filtered day
No cut beats the rule	MOBILE-USER DAY	5	one trained model, seven grading passes
-3.0 from shuffling the hours	MOBILE-USER DAY	6	re-grading the same trained copy
+2.6 at 18-20h	MOBILE-USER DAY	6	the targeted-sampling run, whole day graded
27.4 separation of the derived hours	MOBILE-USER DAY	7	analysis of the kinds of user on the training file
+3.7, first run above the rule	MOBILE-USER DAY	8	routing each kind of user to its own numbers
+4.2 at 1,900 training people	2,000-USER DAY	9	scaling run on the 2,000-person day
-1.6 on stretched days	LEVELLED CORPUS	9	the 200-record levelling run
+4.3, flat from 5,000 to 21,000	ELEVEN-DAY SET	10	four-size scaling notebook, constant training time
The resampling intervals	ELEVEN-DAY SET	10	2,000 paired resamples over the test people
0 of 7 weekday trainings finished	ELEVEN-DAY SET	11	seven-day notebook; the session crashed after 5 h

Rather than reproducing entire configuration files, **Table B.3** isolates eleven critical configuration keys that directly encode substantive methodological decisions argued in the text (such as context evaluation cuts, early stopping patience, transition oversampling ratios, and per-group shrinkage parameters) rather than arbitrary tuning preferences. Altering any of these eleven keys explicitly modifies an empirical claim or scope constraint established in this report.

Table B.3: The settings that encode a decision, not a preference

Key	Value	The decision it encodes
context_fraction	0.8	where each day is cut; part of the ruler (Ch. 4)
max_epochs	20	also the decay horizon: lowering it changes the regime
early_stop_patience	4	the best pass arrives early on this task (Ch. 4)
transition_windows	(4,6, $\times$ 2), (18,20, $\times$ 4)	the <i>guessed</i> hours of Ch. 6; fallback only
group_windows	None	derive the hours from the data, per kind of user (Ch. 7)
transition_chunk_oversample	3.0	show slices containing a move three times more often
group_bands	4 stretches	coarse enough for each stretch to be estimable
group_shrinkage	8.0	a thin stretch must not dominate a profile
group_detect_fraction	past only	no leakage: the kind of user is read from the past
max_steps (scaling)	1,000	the same training time at every size (Ch. 8)
val_subsample (scaling)	250 fixed people	the same stopping rule at every size

C Glossary

The terms that recur from chapter to chapter, each in one or two sentences.

Antenna, mast, cell_id. A *mast* is one physical installation. It carries three or four *antennas* pointing in different directions, and each of those is what the working files call a *cell_id*, for instance UKVCHE102, whose middle letters identify the mast. Mobility is always measured on the mast, because 12.4 out of every 100 back-to-back records change antenna without changing place; what has to be predicted stays the full antenna name.

Cell-switch rate. Out of every hundred back-to-back records of one person, how many land on a different antenna than the one before. Near 0 % for somebody who stays home all day, near 100 % for somebody constantly moving. It is the single property that best predicts how well that person is scored.

Correct answers per hundred. How every result here is expressed. The model gaining 4.3 means it answers correctly 54.5 times out of a hundred where the one-line rule answers correctly 50.2 times, on the same people and moments.

Held-out people. People set aside before training and never shown to the model, on which every reported score is measured.

Hour symbol. A symbol <H00> to <H23> placed before each event. Always handed to the model, never predicted by it, because the current time is known when a prediction is asked for.

Kind of user, or behavioural profile. One of four groups of people, sorted automatically by how often their *mast* changes over four broad stretches of the day, named by increasing mobility: G0 the stay-at-home user, G1 the steady user, G2 the evening-moving user, G3 the morning-moving user.

LoRA, QLoRA. Training methods that freeze the model's original numbers and train a small set of new ones beside them, about 2 in every 100 here. QLoRA additionally stores the frozen numbers in a coarser form.

One-line rule, or persistence baseline. The rule “the next antenna is the last one seen”. No model, no training, and not a constant: 69.2 % on a raw day and 43.1 % on people who genuinely move, because the people differ.

Paired resampling. A test in which the same test people are redrawn at random for both models being compared, so the luck of the draw cancels out. A range containing zero means the two cannot be told apart.

Protocol: last fifth, or whole day. “Last 20 %” grades only the final fifth of a test person's day; “whole day” grades every position of it, which is harder and lowers the one-line rule on the very same people from 45.5 % to 43.1 %.

Record, or event. One antenna name paired with one time. Not a movement: an untouched phone still produces records, because the network asks it to announce itself on a fixed cycle.

Share of the day given, or context fraction. How much of a person's day is handed to the model before it has to produce the rest, 80 % by default. It belongs to the measurement rather than to the model: moving it moves the one-line rule too. Not the model's *context window*, the hard limit of 512 symbols it may look at in one go.

Top-1, top-3, top-5. Whether the antenna the person really visited is the model's first guess, or anywhere among its first three or first five out of the 369 it knows.

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